# Model-Based Calibration of the Tully-Fisher Relation for DESI DR2 Peculiar Velocities

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[DRAFT] We present a model-based calibration of the Tully-Fisher relation (TFR) using the full spiral-galaxy sample from the DESI Data Release 2 (DR2) Peculiar Velocity survey. The DR2 catalog contains 10,200 spiral galaxies drawn from the Siena Galaxy Atlas. We define the TFR phase-space variables $x = \log_{10}(V_{\rm rot}/100\ {\rm km\,s^{-1}})$ and $y = M_R$ (the $r$-band isophotal absolute magnitude at $SB = 26$ mag arcsec$^{-2}$), and fit a two-component Gaussian mixture model (GMM) to the observed ($x$, $y$) distribution to characterize the spiral population. The dominant GMM component (weight $w = 0.935$) defines the orientation of oblique selection cuts in phase space; galaxies passing these cuts together with a redshift window $0.03 < z < 0.1$ constitute the training sample of 5,050 galaxies. We calibrate the TFR by fitting a Stan tophat model to this sample via Hamiltonian Monte Carlo, obtaining slope $\alpha = -6.691 \pm 0.073$, intercept $\beta = -19.245 \pm 0.015$, intrinsic scatter along the velocity axis $\sigma_{{\rm int},x} = 0.013 \pm 0.006$, and intrinsic scatter along the magnitude axis $\sigma_{{\rm int},y} = 0.438 \pm 0.011$. The calibrated model is applied to the full DR2 spiral sample to produce a peculiar velocity catalog (`tophat_catalog.fits`) and a full posterior predictive covariance matrix (`tophat_cov.fits`).

## I. INTRODUCTION

Peculiar velocities — the deviations of galaxy redshifts from pure Hubble flow — encode the gravitational influence of large-scale structure and provide an independent probe of the growth rate of density perturbations, bulk flows, and the Hubble constant [e.g. 4, 5, 18]. The Tully-Fisher relation [TFR; 16] links the intrinsic luminosity of a spiral galaxy to its rotation velocity. Because more luminous spirals rotate faster, a consequence of the dark-matter halos that govern galaxy dynamics [8], the TFR provides a distance estimate independent of redshift: the luminosity is inferred from the rotation velocity, and the distance from the luminosity–flux ratio. Peculiar velocity surveys based on the TFR have been carried out by a number of groups [e.g. 5, 6, 11, 12, 18].

The Dark Energy Spectroscopic Instrument [DESI; **?** ] is now delivering the largest spectroscopic peculiar-velocity dataset to date. The DESI Peculiar Velocity survey [13] targets spiral galaxies, early-type galaxies (Fundamental Plane), and supernova hosts. A first TFR catalog of 1,106 spirals from the DESI Early Data Release [EDR; 1] was presented in Douglass et al. [3], with particular depth in the Coma cluster region. Data Release 1 (DR2) extends the spiral sample to 10,200 galaxies spanning the full DESI footprint, forming the dataset analyzed here.

Traditional TFR calibrations fit a line to the data after applying iterative bias corrections for selection effects and measurement errors [4, 7, 15]. We instead adopt a forward-modeling approach: we write down a probabilistic galaxy model that contains the TFR as its core relationship, and calibrate it by maximizing the likelihood of the observations. This has several advantages over the traditional approach. Measurement uncertainties in both the rotation velocity and magnitude, including the intrinsic scatter of the TFR, enter the likelihood directly without the need for separate bias corrections. The selection function is folded into the likelihood, so no iterative reweighting of the sample is required. The TFR slope, intercept, and intrinsic scatter parameters are all determined in a single joint fit. This modeling approach does require specifying the intrinsic distribution of galaxy rotation velocities; we address this by fitting a Gaussian mixture model to the observed phase space and adopting a uniform (tophat) prior within the selected region (§IV–**??**).

In this paper we present a model-based calibration of the TFR using the DESI DR2 spiral sample. We describe the sample selection criteria (§III), the Stan tophat model and its likelihood (§IV), the GMM-based phase-space selection function (§**??**), the MCMC fit results and diagnostics (§**??**), and the output peculiar velocity catalog (§**??**).

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## II. DATA

The DESI DR2 Peculiar Velocity spiral sample contains 10,200 galaxies drawn from the Siena Galaxy Atlas [SGA; 9], which is based on the DESI Legacy Imaging Surveys DR9 data [2] and covers areas north of $-30°$ declination. The SGA includes galaxies with diameters larger than 20" at the 25th magnitude isophote, denoted as $D(25)$. To be part of the TFR sample, a galaxy must have a Sersic index less than 2, indicating it is likely a spiral galaxy, and an inclination angle greater than 25°, ensuring a significant portion of the galaxy's rotational velocity is visible and the galaxy's major axis is properly aligned.

SGA provides galaxy shape and magnitude information at the $r$-band $SB = 26$ mag arcsec$^{-2}$ isophote, in the form of the photometric axis ratio $b/a$, the position angle PA of the semi-major axis, and the radius $R_{26}$.

The DESI measurements are taken at the galactic center and on both sides along the major axis oriented according to the SGA PA. Rotational velocities were measured at $0.4R_{26}$, the distance at which reliable redshifts could consistently be measured. The redshifts measured at different positions are combined to obtain a redshift $z$ and measured rotational velocity $V_{\mathrm{obs}}$ for each galaxy [3].

SGA and DESI data are combined to infer the rotation velocity $V_{\mathrm{rot}}$. The galaxy inclination angle $i$ is derived from

$$\cos^2 i = \frac{(b/a)^2 - q_0^2}{1 - q_0^2}, \tag{1}$$

where $q_0 = 0.2$ [17]. This is in turn used to convert the observed line-of-sight velocity to the rotation velocity as

$$V_{\mathrm{rot}} = \frac{V_{\mathrm{obs}}}{\sin i}. \tag{2}$$

The analysis variables used throughout this paper are

$$x = \log_{10}\left(\frac{V_{\mathrm{rot}}}{100\ \mathrm{km\,s^{-1}}}\right), \tag{3}$$

$$y = M_R, \tag{4}$$

where $M_R$ is the $r$-band SB26 absolute magnitude computed from the apparent magnitude and the luminosity distance at the observed redshift.

The intrinsic galaxy distribution is redshift-dependent, as illustrated in Figure 1, which shows the average redshift of DR2 spiral galaxies in bins of rotation velocity and absolute magnitude. Intrinsically faint galaxies are preferentially observed at low redshift, reflecting a genuine property of the local universe rather than an observational bias: the SGA catalog is selected on angular size ($D(25) > 20''$) and is therefore not subject to apparent-magnitude selection effects. However, the low-redshift sample is more heavily contaminated by dwarf galaxies, which do not follow the TFR and are thus poorly suited for calibration. We therefore apply a minimum redshift cut for the TFR calibration sample, and caution that TFR-inferred distances for low-redshift galaxies may be more susceptible to contamination by dwarfs that do not follow the TFR. This redshift dependence of the galaxy population also propagates into redshift-dependent covariance in the TFR-inferred peculiar velocities.

## H$\alpha$ REDSHIFT RANGE IN DECAM $r$-BAND

The H$\alpha$ emission line has a rest-frame wavelength of $\lambda_{\mathrm{H}\alpha} = 656.3$ nm. The DECam $r$-band spans approximately $\lambda_{\mathrm{blue}} = 562$ nm to $\lambda_{\mathrm{red}} = 698$ nm at the 50% transmission points.

The observed wavelength of a spectral line is related to its rest-frame wavelength by:

$$\lambda_{\mathrm{obs}} = \lambda_{\mathrm{emit}}\,(1 + z), \tag{5}$$

so the corresponding redshift is:

$$z = \frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{emit}}} - 1. \tag{6}$$

At the blue edge of the $r$-band:

$$z_{\mathrm{blue}} = \frac{562}{656.3} - 1 \approx -0.14, \tag{7}$$

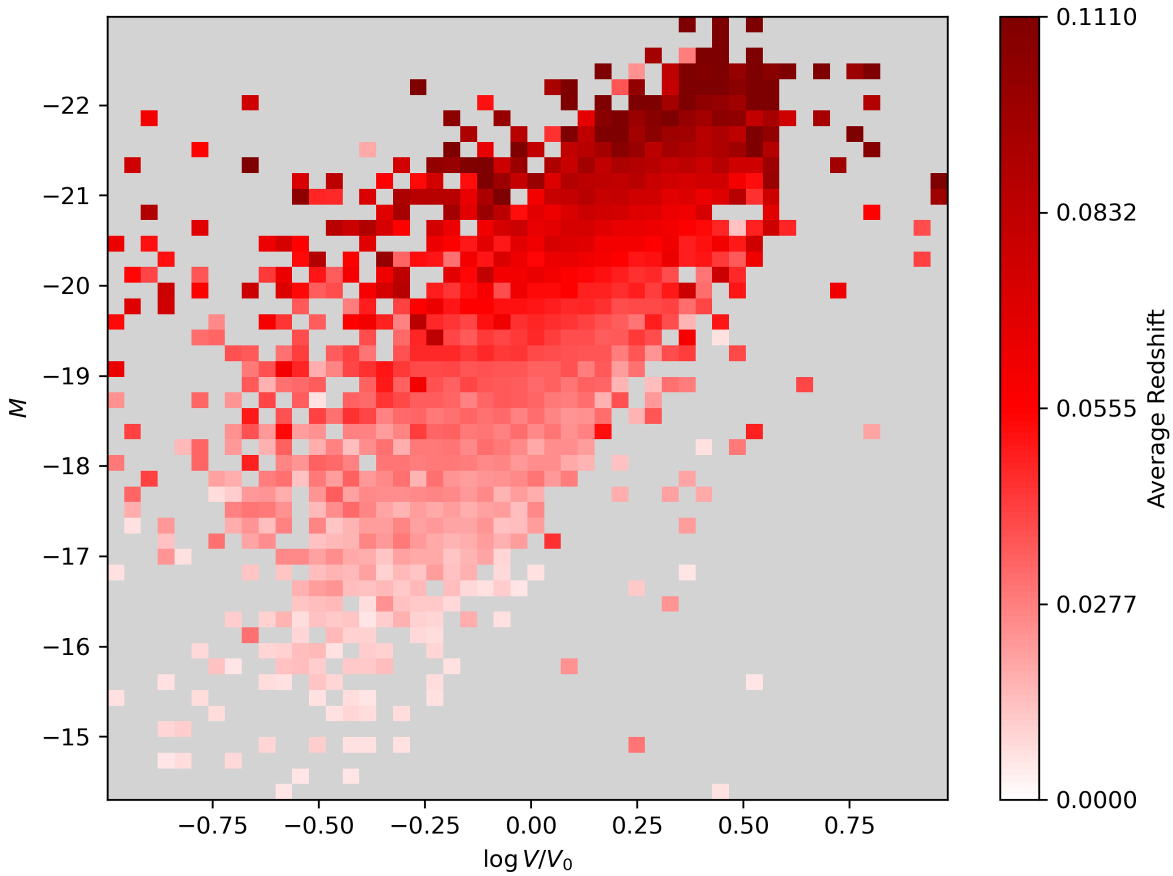


FIG. 1. Average redshift of DR2 spiral galaxies in bins of $\log_{10}(V_{\rm rot}/100\ {\rm km\,s^{-1}})$ and $M_R$. Intrinsically faint, slowly rotating galaxies are preferentially found at low redshift, reflecting the local galaxy population rather than an observational selection effect.

which corresponds to a blueshift and is not physical in a cosmological context.

At the red edge of the $r$-band:

$$z_{\rm red} = \frac{698}{656.3} - 1 \approx 0.064. \tag{8}$$

Since H$\alpha$ at rest ($z = 0$) falls at 656.3 nm, which is already within the $r$-band, the H$\alpha$ line is observable in the DECam $r$-band over the redshift range:

$$0 \lesssim z \lesssim 0.06. \tag{9}$$

## III. SAMPLE SELECTION

Not all galaxies in the DR2 spiral catalog are suitable for TFR calibration. Dwarf and irregular galaxies at the faint end of the luminosity function do not follow the same TFR as normal spirals, while extreme outliers at either end of the phase-space distribution can disproportionately influence the fit. We therefore restrict the analysis to a region $\mathcal{S}$ in the observed $(\hat{x}, \hat{y})$ plane that selects the well-behaved spiral population. The selection process has two phases, with the second region a refinement of the first.

The region $\mathcal{S}$ is defined by two types of cut. First, upper and lower limits on the observed absolute magnitude $\hat{y}$ exclude the faintest galaxies, which depart from the TFR, and the brightest galaxies, which are rare and susceptible to outlier contamination. Second, two diagonal bands oriented approximately parallel to the TFR — and therefore perpendicular to the direction of TFR scatter — bound the sample on either side of the main locus. Cutting parallel to the relation rather than in $\hat{y}$ or $\hat{x}$ alone efficiently encloses the TFR population across the full range of rotation velocities and avoids sculpting the scatter distribution asymmetrically. Together the four boundaries define a parallelogram-shaped region,

$$(\hat{x}, \hat{y}) \in \mathcal{S} \iff \hat{y}_{\min} \leq \hat{y} \leq \hat{y}_{\max} \text{ and } \bar{s}\,\hat{x} + \bar{c}_1 \leq \hat{y} \leq \bar{s}\,\hat{x} + \bar{c}_2, \tag{10}$$

where $\hat{y}_{\min}$ and $\hat{y}_{\max}$ are the magnitude limits, $\bar{s}$ is the slope of the oblique boundaries, $\bar{c}_1 < \bar{c}_2$ are their intercepts, and $d \equiv \bar{c}_2 - \bar{c}_1$ is the perpendicular band width.

The TFR is unknown a priori, so an initial estimate is used to define the slope of the distribution. Variations in the selected slope perturb the galaxies that enter the analysis, but otherwise do not bias the fits: our pipeline recovers the input slopes of mock surveys generated with a consistent model, independent of the initial estimate. The procedures for selecting the selection regions are presented in §V A, after the description of the model.

## IV. MODEL

We construct a probabilistic forward model for the TFR that accounts for measurement uncertainties in both observables and for sample selection within a single likelihood. The full derivation is given in Appendix A; here we state the elements of the model and the expression for the likelihood.

The sample satisfies the selection criteria $(\hat{x}_i, \hat{y}_i) \in \mathcal{S}$. The training sample consists of $N$ galaxies. For a galaxy indexed by $i$, the observables are the measured log-velocity $\hat{x}_i$ with uncertainty $\sigma_{x,i}$ and the observed absolute magnitude $\hat{y}_i$ with uncertainty $\sigma_{y,i}$. Magnitude and log-velocity uncertainties are treated as Gaussian.

The global TFR parameters are the slope $\alpha$, intercept $\beta$, and intrinsic scatter amplitudes $\sigma_{\text{int},x}$ and $\sigma_{\text{int},y}$ along the $x$- and $y$-axes respectively. Each galaxy has a latent on-TFR magnitude $y_{\text{TF},i}$ that places it on the relation. The true absolute magnitude and velocities $x_i$ and $y_i$ have scatter around the TFR,

$$y_i \mid y_{\text{TF},i} \sim \mathcal{N}\big(y_{\text{TF},i},\, \sigma^2_{\text{int},y}\big)\,, \tag{11}$$

$$x_i \mid y_{\text{TF},i} \sim \mathcal{N}\left(\frac{y_{\text{TF},i} - \beta}{\alpha},\, \sigma^2_{\text{int},x}\right). \tag{12}$$

Measurements have Gaussian scatter around the true values:

$$\hat{x}_i \mid x_i \sim \mathcal{N}\big(x_i,\, \sigma^2_{x,i}\big)\,, \tag{13}$$

$$\hat{y}_i \mid y_i \sim \mathcal{N}\big(y_i,\, \sigma^2_{y,i}\big)\,. \tag{14}$$

For a hierarchical model, the underlying distribution of the latent parameter must be specified. We adopt a uniform distribution, $y_{\text{TF},i} \sim \text{Uniform}(y_{\min}, y_{\max})$, with bounds extending slightly beyond the selection window to accommodate scatter at the selection boundary,

$$y_{\min} = \hat{y}_{\min} - \Delta y_{\min}, \qquad y_{\max} = \hat{y}_{\max} + \Delta y_{\max}, \tag{15}$$

where we set $\Delta y_{\min} = -0.5$, $\Delta y_{\max} = 1.0$. The choice of latent model matters only for objects whose latent distribution varies appreciably across the width of the Gaussian measurement error; for objects with small uncertainties the likelihood is insensitive to the model choice. To minimize sensitivity to this choice, we place the selectrion criteria bounds in a region where $p(y_{\text{TF}})$ is expected to have a weak gradient, so that the uniform remains a good local approximation even for objects with broad likelihoods. In this regime the normalization $P(S_i = 1 \mid \theta)$ varies only weakly with small changes to $\Delta y_{\min}$ and $\Delta y_{\max}$ (chosen to be signifcantly larger than the absolute magnitude measurement uncertainties), and the inferred TFR parameters are insensitive to the precise edge placements. This choice is also supported empirically: in mock-catalog tests, the uniform prior yielded more stable parameter recovery than a Gaussian prior, consistent with the relatively flat interior of the galaxy absolute-magnitude distribution.

The above model specifies the likelihood. Given the independence of the galaxy measurements, the total likelihood can be decomposed into the product of per-galaxy terms.

$$\mathcal{L}(\theta; \{\hat{x}_i\}, \{\hat{y}_i\}) = \prod_{i=1}^{N} p(\hat{x}_i, \hat{y}_i \mid \theta, (\hat{x}_i, \hat{y}_i) \in \mathcal{S}). \tag{16}$$

Convolving intrinsic and measurement scatter defines the combined uncertainties

$$\sigma^2_{1,i} = \sigma^2_{x,i} + \sigma^2_{\mathrm{int},x}, \qquad\qquad \sigma^2_{2,i} = \sigma^2_{y,i} + \sigma^2_{\mathrm{int},y}. \tag{17}$$

Because the sample is restricted to $(\hat{x}_i, \hat{y}_i) \in \mathcal{S}$, the per-galaxy density is truncated and renormalized,

$$p(\hat{x}_i, \hat{y}_i \mid \theta, (\hat{x}_i, \hat{y}_i) \in \mathcal{S}) = \frac{\mathcal{L}_i \; \mathbf{1}\{(\hat{x}_i, \hat{y}_i) \in \mathcal{S}\}}{P(S_i = 1 \mid \theta)}. \tag{18}$$

The numerator is the per-galaxy likelihood if there were no sample selection,

$$\mathcal{L}_i = \frac{|\alpha|}{y_{\max} - y_{\min}} \, \mathcal{N}\big(\hat{y}_i;\, \beta + \alpha \hat{x}_i,\, \sigma^2_{\mathrm{tot},i}\big) \left[\Phi\!\left(\frac{y_{\max} - \mu_{*,i}}{\sigma_{*,i}}\right) - \Phi\!\left(\frac{y_{\min} - \mu_{*,i}}{\sigma_{*,i}}\right)\right], \tag{19}$$

where $\Phi$ is the standard normal CDF and

$$\sigma^2_{\mathrm{tot},i} = \alpha^2 \sigma^2_{1,i} + \sigma^2_{2,i}, \tag{20}$$

$$\mu_{*,i} = \frac{(\alpha \hat{x}_i + \beta)\, \sigma^2_{2,i} + \hat{y}_i \, \alpha^2 \sigma^2_{1,i}}{\sigma^2_{\mathrm{tot},i}}, \tag{21}$$

$$\sigma^2_{*,i} = \frac{\alpha^2 \sigma^2_{1,i} \, \sigma^2_{2,i}}{\sigma^2_{\mathrm{tot},i}}. \tag{22}$$

The denominator, i.e. the selection probability, is derived by transforming the $\hat{x}$–$\hat{y}$ to a new coordinate system where the parallelogram transforms to an axis-aligned rectangle. The rectangular region is the intersection of four closed half-planes. The selection probability of a half plane is the standard bivariate-normal CDF. The resulting expression is

$$P(S_i = 1 \mid \theta) = \frac{1}{y_{\max} - y_{\min}} \int_{y_{\min}}^{y_{\max}} I_i(y_{\mathrm{TF}}) \, dy_{\mathrm{TF}}, \tag{23}$$

where $I_i$ is a combination of four bivariate-normal CDF evaluations,

$$\begin{aligned} I_i(y_{\mathrm{TF}}) \equiv\; & \Phi_2\Big(h_i(0; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\max}; y_{\mathrm{TF}});\ -\rho_i\Big) - \Phi_2\Big(h_i(d; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\max}; y_{\mathrm{TF}});\ -\rho_i\Big) \\ & - \Phi_2\Big(h_i(0; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\min}; y_{\mathrm{TF}});\ -\rho_i\Big) + \Phi_2\Big(h_i(d; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\min}; y_{\mathrm{TF}});\ -\rho_i\Big), \end{aligned} \tag{24}$$

with $\Phi_2(h, k; \varrho)$ the standard bivariate-normal CDF, $d \equiv \bar{c}_2 - \bar{c}_1$, and the standardised bounds

$$h_i(t; y_{\mathrm{TF}}) \equiv \frac{\mu_{u,i}(y_{\mathrm{TF}}) - t}{\sigma_{u,i}}, \qquad\qquad k_i(y_\star; y_{\mathrm{TF}}) \equiv \frac{y_\star - y_{\mathrm{TF}}}{\sigma_{2,i}}. \tag{25}$$

Here $\mu_{u,i}$, $\sigma_{u,i}$, and $\rho_i$ are the mean, standard deviation, and correlation of the sheared observable $\hat{u}_i \equiv \hat{y}_i - \bar{s}\hat{x}_i - \bar{c}_1$ conditional on $y_{\mathrm{TF}}$:

$$\mu_{u,i}(y_{\mathrm{TF}}) = y_{\mathrm{TF}} - \bar{s}\, \frac{y_{\mathrm{TF}} - \beta}{\alpha} - \bar{c}_1, \tag{26}$$

$$\sigma^2_{u,i} = \sigma^2_{2,i} + \bar{s}^2 \, \sigma^2_{1,i}, \tag{27}$$

$$\rho_i = \frac{\sigma_{2,i}}{\sigma_{u,i}}. \tag{28}$$

## V. TULLY-FISHER RELATION FIT

### A. Training Set Selection

In this section we describe the procedure for determining the parameters that define the sample selection region $\mathcal{S}$ (Eq. 10), as introduced in §III. The procedure has three steps: (i) a Gaussian Mixture Model (GMM) fit identifies the core spiral population and defines a preliminary selection parallelogram; (ii) a nominal TFR is fit to the galaxies

TABLE I. Fiducial selection parameters for the DR2 training sample.

| Parameter | Value |
|---|---|
| $\hat{y}_{\min}$ | $-21.85$ |
| $\hat{y}_{\max}$ | $-19.00$ |
| $\bar{s}$ | $-6.535$ |
| $\bar{c}_1$ (lower intercept) | $-20.565$ |
| $\bar{c}_2$ (upper intercept) | $-18.021$ |
| $n_{\sigma,\perp}$ | 3.0 |
| $z_{\min}$ | 0.03 |
| $z_{\max}$ | 0.08 |
| Training sample $N$ | 5,050 |

within that parallelogram; and (iii) a residual-profile diagnostic applied to the full catalog refines the magnitude limits to exclude regions where magnitude deviations are significant.

Velocity–magnitude cuts are designed to exclude outliers and sensitivity to the modeling of the latent parameter distribution. We apply a minimum and maximum redshift cuts ($0.03 < z < 0.08$) to reduce sensitivity to peculiar-velocity contamination and potential contamination.

The primary contaminants are dwarf galaxies, which are known not to follow the TFR, and bright galaxies lying on the sharp declining tail of the true luminosity function, where the top-hat model may not yield accurate measurement probabilities. To identify magnitude limits that excise these populations, we apply the nominal TFR to all galaxies and examine the residuals between predicted and observed magnitudes as a function of absolute magnitude. Because the nominal TFR need not be accurate, a smooth linear trend in the residuals is expected and acceptable; we instead look for deviations from this trend, which signal contamination by populations that do not follow the TFR. Galaxies at absolute magnitudes where such deviations appear are excluded from the sample.

The initial TFR is calculated using a subset of galaxies dominated by normal spirals, selected as the core of the distribution via a two-component GMM fit to the ($x$, $y$) distribution of galaxies passing a magnitude pre-filter $-23 < y < -18$ (to remove objects with implausible absolute magnitudes). The fit accounts for per-galaxy measurement noise and rectangular survey truncation at the bright and high-velocity limits of the data. The dominant component represents the TFR spiral population; the secondary component captures outliers and dwarf galaxies.

The best-fit GMM yields a dominant-component weight of $w = 0.935$. The mean of the dominant component is $(\bar{x}, \bar{y}) = (0.171, -20.409)$, with semi-axes [0.064, 0.729] and orientation angle 98.7° measured from the $x$-axis (note: the complementary angle appears in plots due to magnitude-axis orientation conventions). The GMM $1\sigma$ and $3\sigma$ ellipses overlaid on the phase-space scatter are shown in Fig. 2.

A preliminary selection parallelogram is defined by the $3\sigma$ ellipse of the dominant GMM component. The slope $\bar{s}$ is fixed by the orientation of the ellipse, and $\bar{c}_1$, $\bar{c}_2$ are set at $n_{\sigma,\perp} = 3$ perpendicular $\sigma$ from the ellipse center. Values of $\hat{y}_{\min}$ and $\hat{y}_{\max}$ are taken from the $y$-extent of the $3\sigma$ ellipse. The redshift limits used in the GMM fit are maintained. The selected parallelogran defining the initial training region is shown in Fig. 2.

The nominal TFR is given by the maximum likelihood estimator of the model fit to the data within the parallelogram, yielding a slope of $-6.8346$, an intercept of $-19.2215$, an intrinsic scatter in $x$ of $\sigma_{\mathrm{int},x} = 6.132 \times 10^{-2}$, and an intrinsic scatter in $y$ of $\sigma_{\mathrm{int},y} = 0.0991$.

For the full sample of galaxies with $0.03 < z < 0.08$, we predict absolute magnitudes using the nominal TFR and compute the residual $\Delta M = \langle M_{\mathrm{pred}} \rangle - M_{\mathrm{obs}}$ for every object. We then calculate the weighted mean residual and its uncertainty in equal-occupancy bins of $M_R$. Since the nominal TFR slope is only approximate, we expect a linear trend in this residual profile: an imperfect slope produces a systematic over- or under-prediction that varies smoothly with magnitude. Contamination by populations that do not follow the TFR — most notably dwarf galaxies — produces deviations from this linear trend at the affected magnitude ranges. The magnitude limits $\hat{y}_{\min}$ and $\hat{y}_{\max}$ are set at the values beyond which the residual profile departs from the linear trend, identified by visual inspection. The residual profile for the fiducial selection is shown in Fig. 3; the linear trend is consistent with the data across the selected range. The fiducial selection parameters for the DR2 run are listed in Table I and shown with respect to the full sample in Figure 4.

### B. MCMC Fit

The model is fit to the training sample using Hamiltonian Monte Carlo as implemented in Stan [14] via the CmdStan interface. We run 4 chains, each with 500 warm-up and 1000 sampling iterations. Convergence diagnostics are satisfactory for all model parameters: $\hat{R} = 1.0$ and bulk effective sample sizes are reported in Table II. Stan's

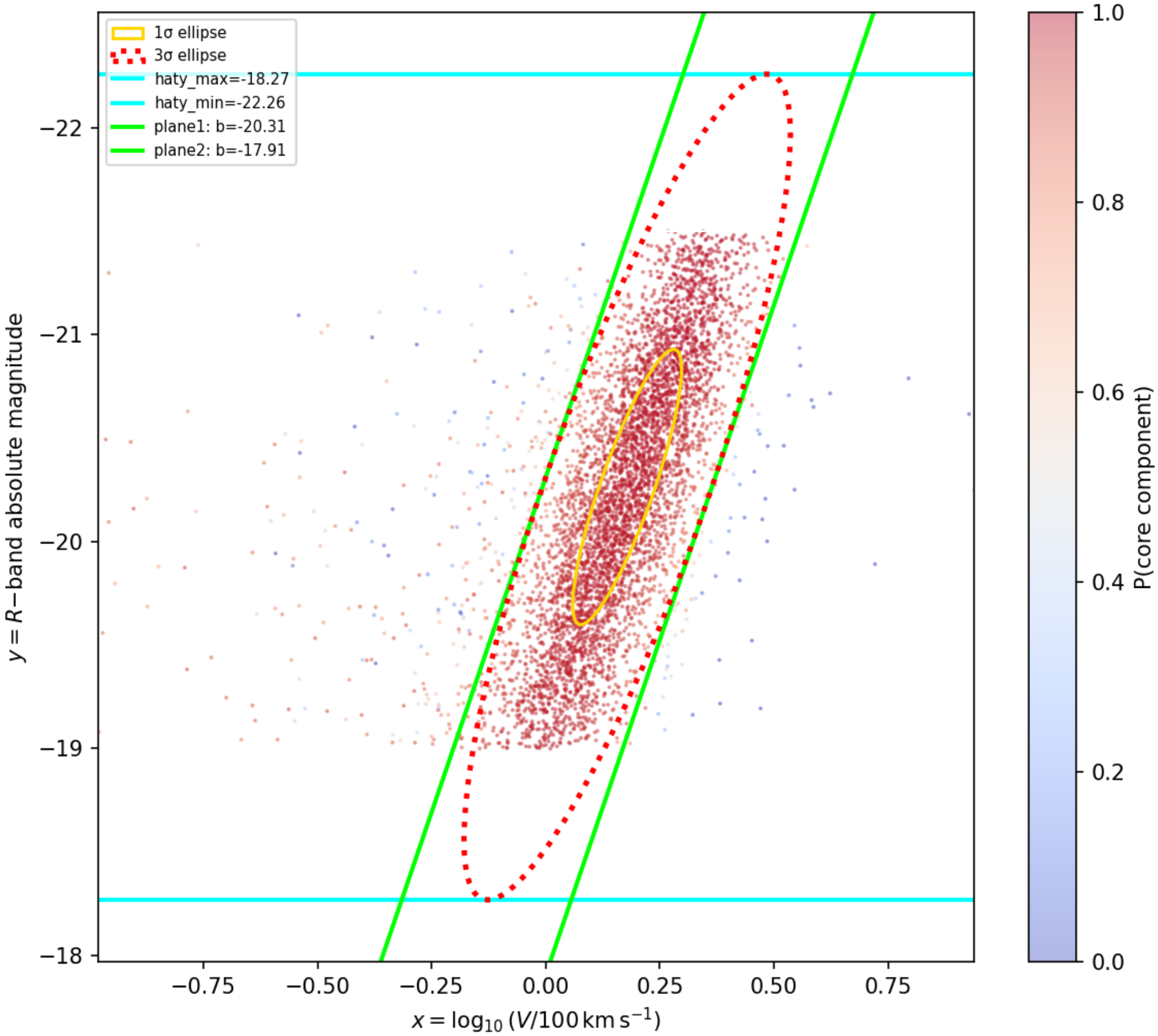


FIG. 2. Distribution of DR2 spiral galaxies used for the initial training set selection in ($x$, $y$) phase space, where $x = \log_{10}(V_{\rm rot}/100\ \mathrm{km\,s^{-1}})$ and $y = M_R$. The two GMM ellipses (at $1\sigma$ and $3\sigma$) are overplotted; the dominant component (weight $w = 0.935$). The parallelogram defined by the extrema of the $3\sigma$ ellipse bound the sample used to determine the initial TFR estimate.

`diagnose` utility confirms no divergent transitions, no treedepth saturation, and satisfactory E-BFMI across all four chains. The posterior 68% confidence intervals for the global TFR parameters are given in Table II. The full posterior corner plot is shown in Fig. 5.

## VI. ABSOLUTE MAGNITUDE PREDICTIONS FROM THE TFR

For all galaxies in the initial DESI DR2 Peculiar Velocity sprial sample, the TFR determined in §V is used to infer their absolute magnitudes. The subset of galaxies satisfying the magnitude–velocity selection criteria of the training sample, with no restriction on redshift is referred to as the `MAIN` sample. Nearly all galaxies in the `MAIN` sample are part of the training set; nevertheless, we treat each as if it were out-of-sample. This approximation is justified on two grounds: no single galaxy has a significant influence on the TFR fit, and accounting for the covariance between the data and the fitted parameters would substantially complicate the analysis.

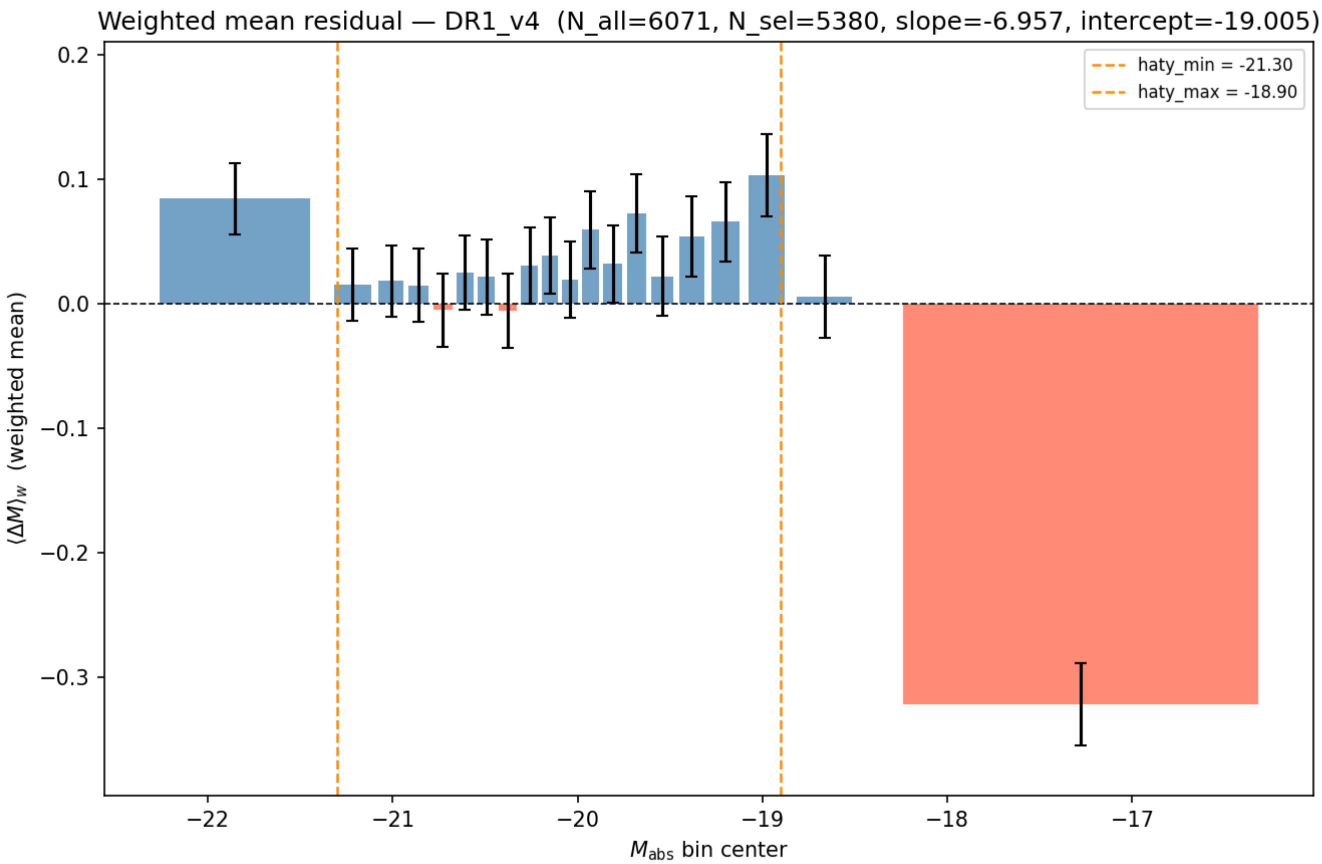


FIG. 3. Weighted-mean residual $\Delta M = \langle M_{\rm pred}\rangle - M_{\rm obs}$ and pull profile (normalized by bin uncertainty) as a function of $M_R$ for all catalog objects evaluated at the MLE parameters. Horizontal lines show the $\hat{y}_{\rm min}$ and $\hat{y}_{\rm max}$ range chosen to select galaxies with a self-consistent TFR.

TABLE II. Posterior summary for the DR2 tophat model fit. Mean and StdDev are computed from the full MCMC chains; 5%, 50%, and 95% are posterior quantiles; $\mathrm{ESS_{bulk}}$ is the bulk effective sample size; $\hat{R}$ is the rank-normalized split potential scale reduction factor.

| Parameter | Mean | StdDev | 5% | 50% | 95% | $\mathrm{ESS_{bulk}}$ | $\hat{R}$ |
|---|---|---|---|---|---|---|---|
| slope $\alpha$ | $-6.691$ | 0.073 | $-6.816$ | $-6.689$ | $-6.573$ | 1178 | 1.0 |
| intercept $\beta$ | $-19.245$ | 0.015 | $-19.270$ | $-19.246$ | $-19.219$ | 1301 | 1.0 |
| $\sigma_{\mathrm{int},x}$ | 0.013 | 0.006 | 0.003 | 0.014 | 0.022 | 875 | 1.0 |
| $\sigma_{\mathrm{int},y}$ | 0.438 | 0.011 | 0.420 | 0.439 | 0.456 | 1414 | 1.0 |

The predicted absolute magnitude for each galaxy is computed via the posterior predictive distribution derived in §B. For a galaxy with measured rotation-velocity proxy $\hat{x}_\star$ and uncertainty $\sigma_{x,\star}$, we adopt a uniform (top-hat) prior on the latent Tully–Fisher magnitude $y_{\mathrm{TF},\star}$ over the selection interval $[y_{\min}, y_{\max}]$. Marginalizing over the model parameters $\theta$ via the MCMC posterior yields the posterior predictive distribution

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \{\hat{x}_i, \hat{y}_i\}) \approx \frac{1}{M}\sum_{m=1}^{M} p\Big(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta^{(m)}\Big)\,, \tag{29}$$

where $\{\theta^{(m)}\}_{m=1}^{M}$ are draws from the MCMC posterior. Each single-draw predictive is given by the closed form

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \mathcal{N}\big(y_\star \mid \mu_L,\, \sigma_Z^2\big)\, \frac{\Phi\Big(\frac{y_{\max} - \tilde{\mu}(y_\star)}{\tilde{\sigma}}\Big) - \Phi\Big(\frac{y_{\min} - \tilde{\mu}(y_\star)}{\tilde{\sigma}}\Big)}{Z_L}, \tag{30}$$

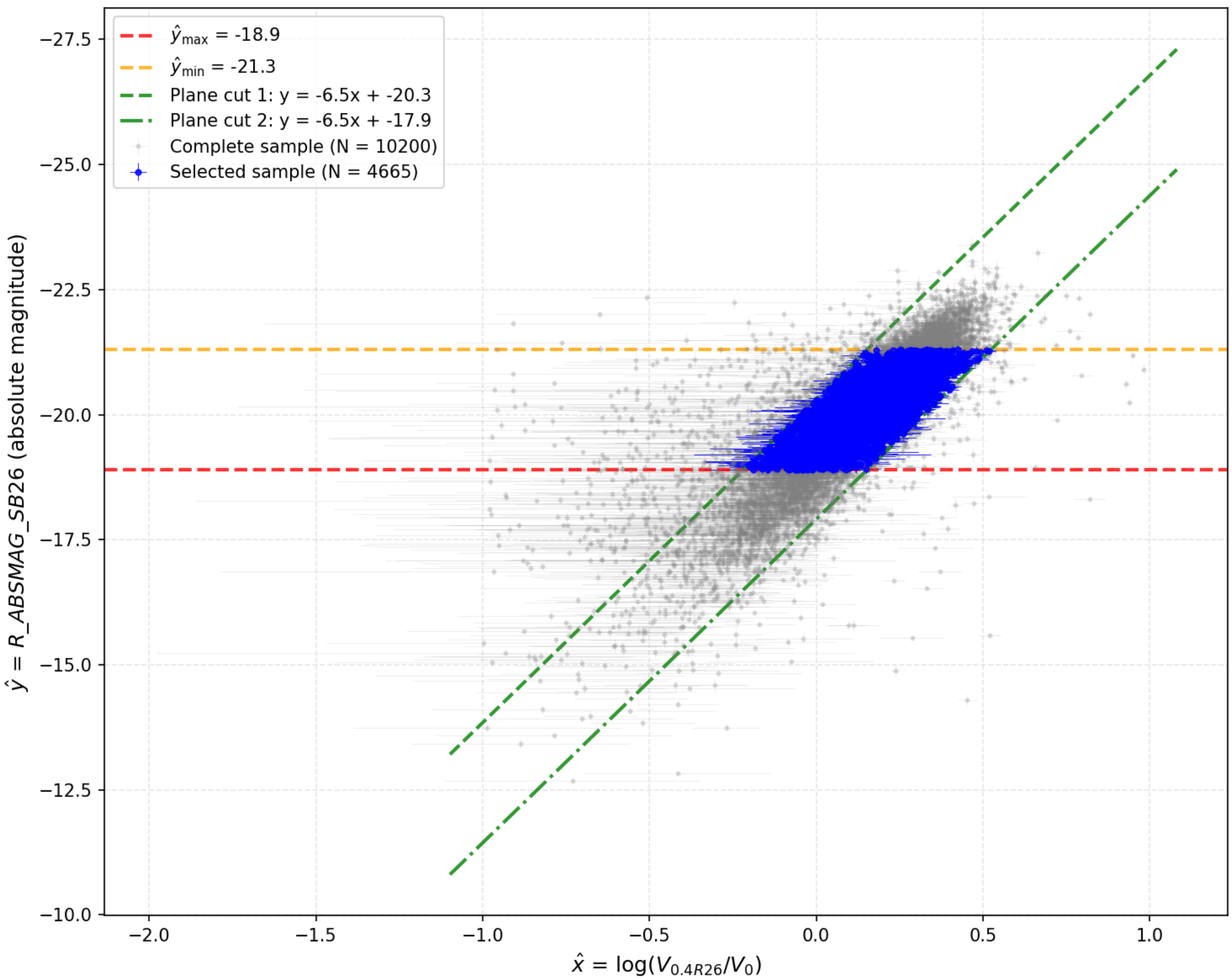


FIG. 4. Full DR2 spiral sample (grey) and training sample (colored points) in $(x, y)$ phase space. The oblique selection boundaries and magnitude limits are indicated. The training sample contains 5,050 galaxies.

where $\Phi(\cdot)$ is the standard normal CDF and $\sigma_{1,\star}^2 = \sigma_{x,\star}^2 + \sigma_{\mathrm{int},x}^2$ as in Eq. (17),

$$\mu_L = \beta + \alpha \hat{x}_\star, \qquad \sigma_Z^2 = \sigma_L^2 + \left(\sigma_{\mathrm{int},y}^{(m)}\right)^2, \qquad \sigma_L = |\alpha|\, \sigma_{1,\star}, \tag{31}$$

the truncation normalization is

$$Z_L = \Phi\!\left(\frac{y_{\max} - \mu_L}{\sigma_L}\right) - \Phi\!\left(\frac{y_{\min} - \mu_L}{\sigma_L}\right), \tag{32}$$

and the convolution quantities arising from marginalizing over $y_{\mathrm{TF},\star}$ are

$$\tilde{\sigma}^2 = \left(\frac{1}{\sigma_L^2} + \frac{1}{\sigma_{\mathrm{int},y}^2}\right)^{-1}, \qquad \tilde{\mu}(y_\star) = \tilde{\sigma}^2 \left(\frac{\mu_L}{\sigma_L^2} + \frac{y_\star}{\sigma_{\mathrm{int},y}^2}\right). \tag{33}$$

The reported magnitude and its uncertainty for each galaxy are the mean and standard deviation of the mixture distribution in Eq. (29). The full posterior predictive covariance across galaxies is obtained via the law of total covariance. Conditional on the TFR parameters $\theta$, galaxy predictions are independent, so off-diagonal covariance arises solely from the shared uncertainty in $\theta$. For each MCMC draw $\theta^{(m)}$, define the per-draw conditional mean

$$\mu_i^{(m)} \;=\; \mathbb{E}\left[y_i \mid \hat{x}_i, \sigma_{1,i}^{(m)}, \theta^{(m)}\right], \tag{34}$$

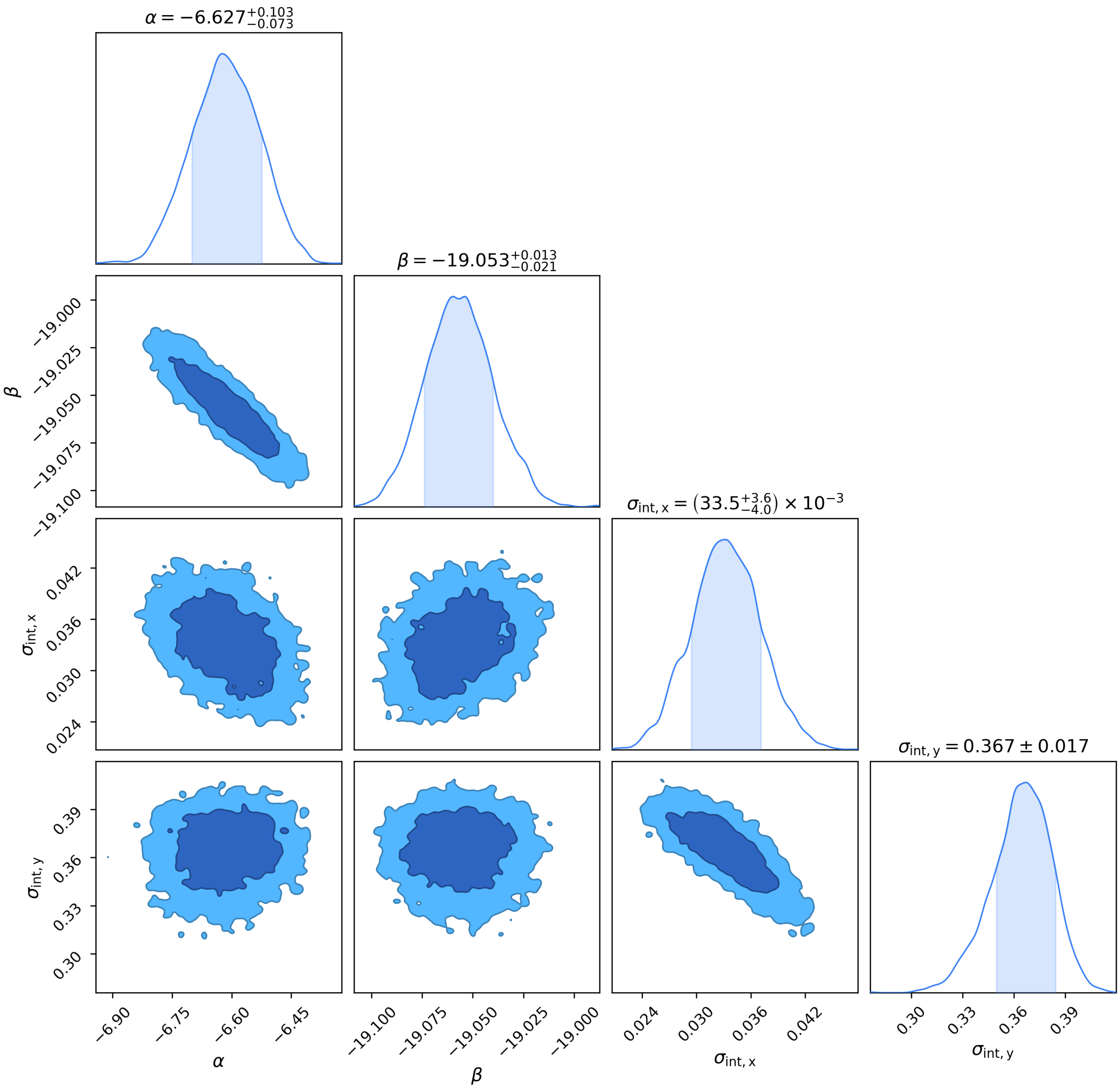


FIG. 5. Corner plot of the posterior distribution for the four global TFR parameters from the DR2 tophat model fit. Contours show the $1\sigma$ and $2\sigma$ confidence regions.

which is the truncated-normal mean of $y_{\mathrm{TF},i}$ given $\hat{x}_i$ and $\theta^{(m)}$ (Eq. (B20)), and the per-draw predictive variance

$$v_i^{(m)} \;=\; \mathrm{Var}\Big(y_{\mathrm{TF},i} \mid \hat{x}_i, \theta^{(m)}\Big) \;+\; \Big(\sigma_{\mathrm{int},y}^{(m)}\Big)^2, \tag{35}$$

including both flavors of intrinsic scatter: $\sigma_{\mathrm{int},y}$ enters additively, while $\sigma_{\mathrm{int},x}$ enters the truncated-normal width through $\sigma_L = |\alpha|\,\sigma_{1,\star}$, where $\sigma_{1,\star}^2 = \sigma_{x,\star}^2 + \sigma_{\mathrm{int},x}^2$. The diagonal of the predictive covariance then decomposes into three contributions: calibration variance ($\mathrm{Cov}_\theta$ of the conditional means), expected predictive variance ($\mathbb{E}_\theta[v_i^{(m)}]$, encompassing intrinsic scatter and truncated-normal variance), and measurement noise $\sigma_{y,i}^2$ in $\hat{y}_i$, where $\bar{y}_i = \frac{1}{M}\sum_m \mu_i^{(m)}$ is the posterior predictive mean. Off-diagonal elements follow from the calibration-variance term alone, exploiting conditional independence. The full posterior predictive covariance matrix for a subsample of galaxies ordered by

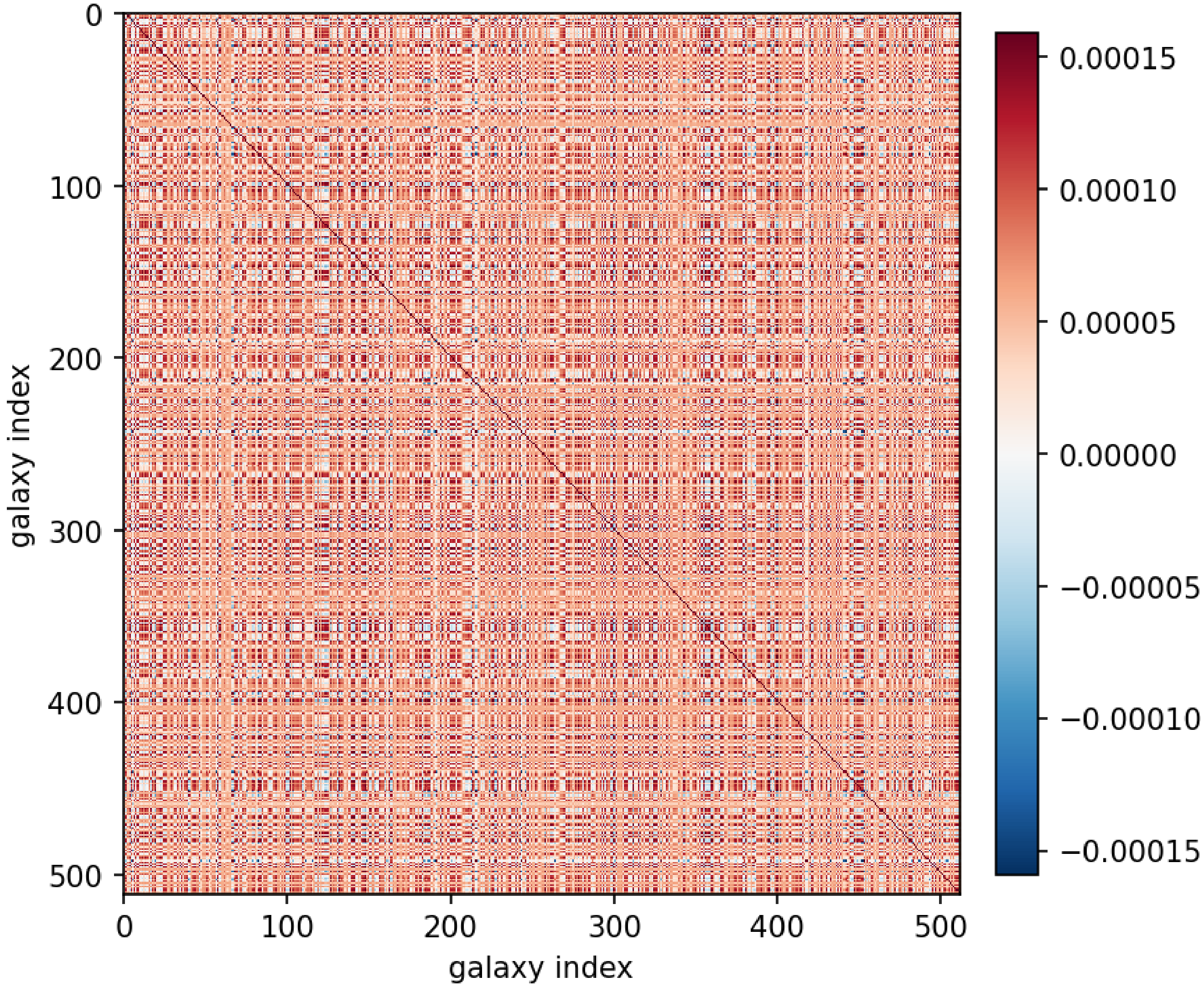


FIG. 6. Visualization of the absolute magnitude posterior predictive covariance matrix for a subsample of galaxies ordered by redshift. Off-diagonal correlations arise from the shared TFR calibration parameters. The diagonal for the matrix of the `MAIN` sample is not clearly visible at the resolution of this plot.

redshift is visualized in Fig. 6.

## VII. PECULIAR VELOCITIES/MAGNITUDES

Peculiar velocities are manifest as peculiar magnitudes, i.e. deviations between observed and TFR-predicted absolute magnitudes. These residuals $\hat{y}_{\mathrm{pred}} - \hat{y}_{\mathrm{obs}}$ are calculated for the DR2 PV spiral sample.

We define a grid in $(x,\, y)$ phase space and compute the mean residual for each element. Figure 7 shows this residual grid, both for the full sample and for the `MAIN` sample.

As discussed in §V A, the magnitude limits $\hat{y}_{\mathrm{min}}$ and $\hat{y}_{\mathrm{max}}$ were chosen to exclude regions where the TFR fit is not self-consistent, by examining the combined magnitude residuals in slices of redshift. The left panel of Fig. 7 illustrates why this is necessary: for galaxies brighter than $M_R = -19$, the zero-residual locus follows the TFR slope, with residuals increasing monotonically with distance from it, consistent with Gaussian scatter and measurement uncertainties. For galaxies fainter than $M_R = -19$, however, the zero-residual locus deviates from this slope, indicating a population — likely dwarf galaxies — that does not follow the TFR. A similar but more sublte deviation is observed at the bright end, which could be due to galaxy outliers or limitations of the top-hat model in this regime. The right panel of Fig. 7

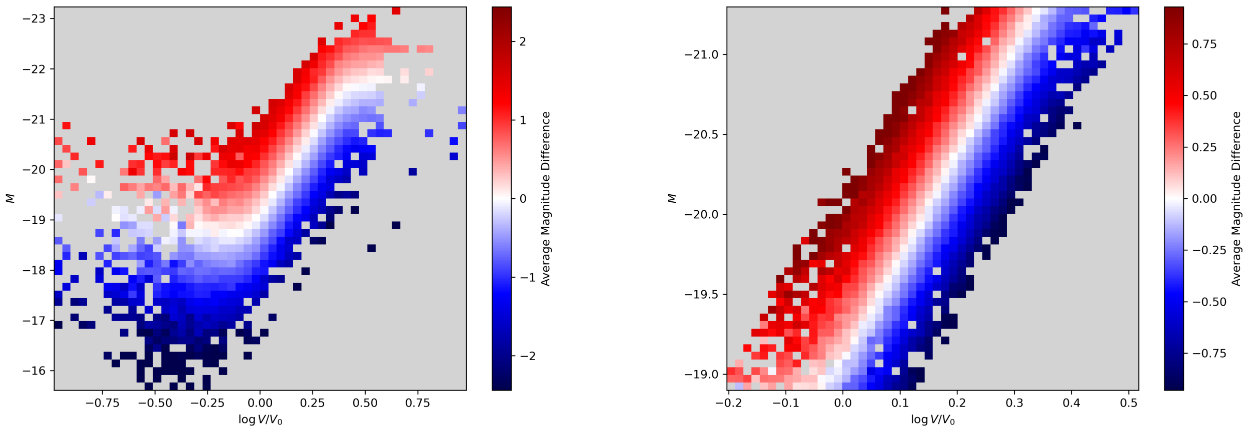


FIG. 7. Mean prediction residual $\hat{y}_{\rm pred} - \hat{y}_{\rm obs}$ binned on a grid in ($x$, $y$) phase space for the training sample. Left: the DR2 PV spiral, Right: the `MAIN` sample. The residuals are consistent with zero, with no evidence for systematic trends.

shows that for the `MAIN` sample, for which selection criteria are applied (but no redshift restriction), the residuals are consistent with zero across the full phase space, with no evidence for a non-linear TFR or magnitude-dependent bias. There is a slight asymmetry in the residuals, as the sample selection parallelogram is not perfectly centered or aligned with the TFR fit to the training sample and used in the selection of the final sample.

The importance of redshift is demonstrated in Fig. 8, which shows the residuals as a function of redshift for both the full sample and the `MAIN` sample. For the full sample, there is a clear redshift dependence in the residuals, with a systematic trend that deviates from zero at low and high redshifts. This indicates that the TFR fit is not self-consistent across the full redshift range, likely due to contamination by non-TFR-following populations (e.g., dwarf galaxies) at certain redshifts. In contrast, for the `MAIN` sample, which is selected based on the criteria defined in §V A, the residuals are consistent with zero across the entire redshift range, at least to first-order, demonstrating that the selection criteria mitigate redshift-dependent systematic effects.

### A. Data Products

The catalog is provided as `tophat_catalog.fits`, which augments the input FITS table with five new columns:

- `MU_TF`: the TFR-inferred distance modulus $\mu_{TF} = m_{R,\rm corr} - \langle M_* \rangle_{\rm pred}$, where $m_{R,\rm corr}$ is the corrected apparent $R$-band magnitude and $\langle M_* \rangle_{\rm pred}$ is the posterior predictive mean absolute magnitude from the TFR model (float32).
- `MU_ERR`: the combined uncertainty $\sigma_{\mu_{TF}} = \sqrt{\sigma^2_{m,\rm corr} + \sigma^2_{\rm pred}}$, where $\sigma_{m,\rm corr}$ is the observational magnitude uncertainty and $\sigma_{\rm pred}$ is the posterior predictive standard deviation (float32).
- `LOGDIST`: the log-distance ratio $\eta = 0.2\,(\mu_z - \mu_{TF})$, where $\mu_z = m_R - M_R(z)$ is the CMB-frame distance modulus derived from the observed redshift. Equivalently, $\eta = \log_{10}(d_H/d_{TF})$, the base-10 logarithm of the ratio of the Hubble distance to the TFR distance (float32).
- `LOGDIST_ERR`: the uncertainty on `LOGDIST`, $\sigma_\eta = 0.2\,\sigma_{\mu_{TF}}$ (float32).
- `MAIN`: boolean flag (`True`/`False`) indicating whether the galaxy satisfies the phase-space selection cuts (absolute magnitude range and oblique TFR-plane boundaries defined in `config.json`), evaluated without a redshift restriction. Only `MAIN`=`True` galaxies are included in `tophat_cov.fits`.

The posterior predictive covariance matrix is provided as `tophat_cov.fits`. The file contains a single $N \times N$ float32 image extension, where $N$ is the number of `MAIN`-flagged galaxies. Rows and columns are ordered to match the `MAIN`=`True` rows of `tophat_catalog.fits` in their original catalog order. Entry $(g, g')$ gives $\mathrm{Cov}(\mu^{(g)}_{TF}, \mu^{(g')}_{TF})$, the posterior predictive covariance of the TF distance moduli; diagonal entries additionally include the per-galaxy

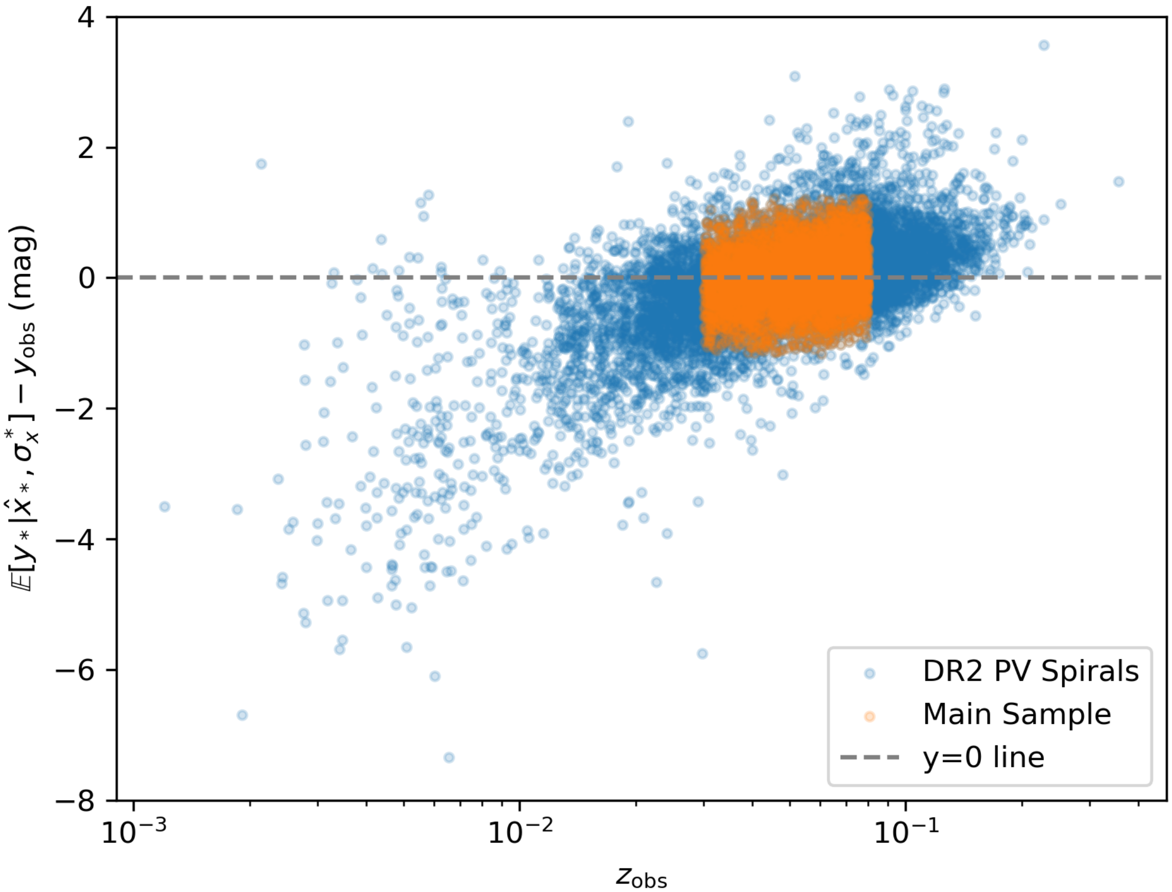


FIG. 8. TFR residuals as a function of observed redshift for the input and `MAIN` samples.

observational magnitude variance $\sigma^2_{m,\mathrm{obs}}$. This matrix encodes the correlated uncertainties arising from the shared TFR calibration parameters and is required for cosmological analyses that propagate correlated peculiar velocity errors. The matrix structure is visualized in Fig. 6.

## VIII. CONCLUSIONS

We have presented a model-based calibration of the Tully-Fisher relation using the full spiral-galaxy sample from the DESI DR2 Peculiar Velocity survey. The key results are as follows.

- A two-component GMM fit to the DR2 phase space identifies the dominant spiral TFR population (weight $w = 0.935$) and defines the orientation of oblique selection cuts. The fiducial selection yields a training sample of 5,050 galaxies with a flat pull profile confirming no selection-induced bias.

- The Stan tophat model, which places a uniform prior on the latent log-velocity within the observed selection region, correctly accounts for the complex oblique selection boundary without the need for explicit bias corrections.

- MCMC posterior parameters are slope $\alpha = -6.691 \pm 0.073$, intercept $\beta = -19.245 \pm 0.015$, and intrinsic scatter $\sigma_{\mathrm{int},x} = 0.013 \pm 0.006$, $\sigma_{\mathrm{int},y} = 0.438 \pm 0.011$. Residual diagnostics in both phase space and redshift show no evidence for systematic trends.

- The calibrated model is applied to produce a peculiar velocity catalog (`tophat_catalog.fits`) and a posterior predictive covariance matrix (`tophat_cov.fits`) for the full MAIN sample.

This work is distinguished from other approaches in that it fits a probabilistic TFR model rather than fitting a line directly to the data. The key distinction is that parameter estimation is based on the model likelihood, whereas other approaches optimize a reasonable but essentially arbitrary merit function. A feature of the TFR model is that it must specify the latent parameter distribution, and we find that the results are sensitive to this choice. An advantage of our approach is that the model can be applied to independent datasets in a self-consistent manner.

Peculiar velocity measurements from the MAIN catalog, including the study of large-scale flows and comparison with existing peculiar velocity compilations, will be presented in a follow-up paper.


## ACKNOWLEDGMENTS

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The authors are honored to be permitted to conduct scientific research on Iolkam Du'ag (Kitt Peak), a mountain with particular significance to the Tohono O'odham Nation.


## Appendix A: Fitting the Tully-Fisher Relation

Consider a simple model in which peculiar velocities are not included and the background cosmology is known. In this case observed redshifts can be converted to distances and observed magnitudes can be converted to absolute magnitudes.

### 1. Data

**$N$:** Total number of galaxies.

**$\hat{x}_i$, $\sigma_{x,i}$:** Observed $\log\left(\frac{V_{\rm rot}}{V_0}\right)$ and uncertainty for galaxy $i$.

**$\hat{y}_i$, $\sigma_{y,i}$:** Observed absolute magnitude and uncertainty for galaxy $i$.

The velocity uncertainties are normal in log-space, which is probably not entirely accurate but may be sufficient for our purposes. The benefit of this model is that the likelihood for each galaxy can be computed in closed form, which makes inference tractable and efficient.

### 2. Parameters

**$\alpha$, $\beta$:** Slope and intercept of the Tully–Fisher relation.

**$\sigma_{\mathbf{int},x}$, $\sigma_{\mathbf{int},y}$:** Intrinsic scatter in the $x$-, $y$-directions.

**$y_{\mathbf{TF},i}$:** Latent "on-relation" absolute magnitude for galaxy $i$.

**$x_i$, $y_i$:** Latent true $x$, $y$ for galaxy $i$.

**$\Xi$:** Parameters that describe the distribution of $y_{\rm TF}$.

### 3. Model

Each galaxy has a latent parameter that determines its TF properties. The galaxy absolute magnitude and rotation velocity are drawn independently and probabilistically based on this parameter using the TF relation. Measurement errors are independent and normally distributed. Specifically, for each galaxy $i$:

$$y_i \mid y_{\mathrm{TF},i} \sim \mathcal{N}(y_{\mathrm{TF},i}, \sigma^2_{\mathrm{int},y}), \tag{A1}$$

$$x_i \mid y_{\mathrm{TF},i}, \alpha, \beta \sim \mathcal{N}\left(\frac{y_{\mathrm{TF},i} - \beta}{\alpha}, \sigma^2_{\mathrm{int},x}\right). \tag{A2}$$

The latent parameter $y_{\mathrm{TF},i}$ has an underlying distribution; we consider either a top-hat

$$y_{\mathrm{TF},i} \sim \mathrm{Uniform}(y_{\min}, y_{\max}) \tag{A3}$$

or a Normal

$$y_{\mathrm{TF},i} \sim \mathcal{N}(\mu_{y_{\mathrm{TF}}}, \tau^2) \tag{A4}$$

distribution. Thus $\Xi \equiv \{y_{\min}, y_{\max}\}$ or $\{\mu_{y_{\mathrm{TF}}}, \tau\}$.

### 4. Complete Sample

For a complete sample

$$\hat{x}_i \mid x_i \sim \mathcal{N}(x_i, \sigma^2_{x,i}), \tag{A5}$$

$$\hat{y}_i \mid y_i \sim \mathcal{N}(y_i, \sigma^2_{y,i}). \tag{A6}$$

Let $\theta \equiv \left(\alpha, \beta, \sigma_{\mathrm{int},x}, \sigma_{\mathrm{int},y}, \Xi\right)$. The total likelihood is

$$\mathcal{L}(\theta; \{\hat{x}_i\}, \{\hat{y}_i\}) = \prod_{i=1}^{N} \mathcal{L}_i(\theta; \hat{x}_i, \hat{y}_i), \tag{A7}$$

where the likelihood for an individual galaxy is

$$\mathcal{L}_i = p(\hat{x}_i, \hat{y}_i \mid \theta) \tag{A8}$$

$$= \int dx_i\, dy_i\, dy_{\mathrm{TF}}\; p(\hat{x}_i \mid x_i)\, p(\hat{y}_i \mid y_i)\, p(x_i \mid y_{\mathrm{TF}}, \alpha, \beta)\, p(y_i \mid y_{\mathrm{TF}})\, p(y_{\mathrm{TF}}). \tag{A9}$$

Convolving intrinsic and measurement scatter gives

$$\sigma^2_{1,i} = \sigma^2_{x,i} + \sigma^2_{\mathrm{int},x}, \tag{A10}$$

$$\sigma^2_{2,i} = \sigma^2_{y,i} + \sigma^2_{\mathrm{int},y}, \tag{A11}$$

so that

$$\hat{x}_i \mid y_{\mathrm{TF},i} \sim \mathcal{N}\left(\frac{y_{\mathrm{TF},i} - \beta}{\alpha}, \sigma^2_{1,i}\right), \tag{A12}$$

$$\hat{y}_i \mid y_{\mathrm{TF},i} \sim \mathcal{N}(y_{\mathrm{TF},i}, \sigma^2_{2,i}), \tag{A13}$$

and therefore the per-galaxy likelihood can be written as

$$\mathcal{L}_i = \int dy_{\mathrm{TF}}\; \mathcal{N}\left(\hat{x}_i; \frac{y_{\mathrm{TF}} - \beta}{\alpha}, \sigma^2_{1,i}\right)\, \mathcal{N}\left(\hat{y}_i; y_{\mathrm{TF}}, \sigma^2_{2,i}\right)\, p(y_{\mathrm{TF}}). \tag{A14}$$

*a. Top-Hat Model for $y_{TF,i}$*

Let $y_{\mathrm{TF},i} \sim \mathrm{Uniform}(y_{\min}, y_{\max})$, i.e.,

$$p(y_{\mathrm{TF},i}) = \begin{cases} \frac{1}{y_{\max}-y_{\min}}, & y_{\min} \le y_{\mathrm{TF},i} \le y_{\max}, \\ 0, & \text{otherwise.} \end{cases} \tag{A15}$$

Then

$$\mathcal{L}_i = \frac{1}{y_{\max} - y_{\min}} \int_{y_{\min}}^{y_{\max}} dy_{\mathrm{TF}}\, \mathcal{N}\left(\hat{x}_i; \frac{y_{\mathrm{TF}} - \beta}{\alpha}, \sigma_{1,i}^2\right) \mathcal{N}\left(\hat{y}_i; y_{\mathrm{TF}}, \sigma_{2,i}^2\right) \tag{A16}$$

$$= \frac{|\alpha|}{y_{\max} - y_{\min}} \mathcal{N}\left(\hat{y}_i; \beta + \alpha\hat{x}_i,\ \sigma_{\mathrm{tot},i}^2\right) \left[\Phi\left(\frac{y_{\max} - \mu_{*,i}}{\sigma_{*,i}}\right) - \Phi\left(\frac{y_{\min} - \mu_{*,i}}{\sigma_{*,i}}\right)\right], \tag{A17}$$

where $\Phi$ is the standard normal CDF and

$$\sigma_{\mathrm{tot},i}^2 = \alpha^2\sigma_{1,i}^2 + \sigma_{2,i}^2, \tag{A18}$$

$$\mu_{*,i} = \frac{(\alpha\hat{x}_i + \beta)\,\sigma_{2,i}^2 + \hat{y}_i\,\alpha^2\sigma_{1,i}^2}{\sigma_{\mathrm{tot},i}^2}, \tag{A19}$$

$$\sigma_{*,i}^2 = \frac{\alpha^2\sigma_{1,i}^2\sigma_{2,i}^2}{\sigma_{\mathrm{tot},i}^2}. \tag{A20}$$

*b. Normal Model for $y_{TF,i}$*

Let $y_{\mathrm{TF},i} \sim \mathcal{N}(\mu_{y_{\mathrm{TF}}}, \tau^2)$. Then

$$\mathcal{L}_i = \int_{-\infty}^{\infty} dy_{\mathrm{TF}}\, \mathcal{N}\left(y_{\mathrm{TF}}; \mu_{y_{\mathrm{TF}}}, \tau^2\right) \mathcal{N}\left(\hat{x}_i; \frac{y_{\mathrm{TF}} - \beta}{\alpha}, \sigma_{1,i}^2\right) \mathcal{N}\left(\hat{y}_i; y_{\mathrm{TF}}, \sigma_{2,i}^2\right) \tag{A21}$$

$$= \mathcal{N}_2\left(\begin{bmatrix}\hat{x}_i \\ \hat{y}_i\end{bmatrix}; \begin{bmatrix}\dfrac{\mu_{y_{\mathrm{TF}}} - \beta}{\alpha} \\ \mu_{y_{\mathrm{TF}}}\end{bmatrix}, \Sigma_i\right), \tag{A22}$$

where

$$\Sigma_i = \begin{bmatrix} \dfrac{\tau^2}{\alpha^2} + \sigma_{1,i}^2 & \dfrac{\tau^2}{\alpha} \\ \dfrac{\tau^2}{\alpha} & \tau^2 + \sigma_{2,i}^2 \end{bmatrix}, \qquad \alpha \neq 0. \tag{A23}$$

## 5. Sample Selection

*a. DESI TF Sample Selection*

In practice, sample selection can bias the analysis because the probability that an object enters the final sample need not be independent of its measured properties. A classic example is Malmquist bias: in a magnitude-limited survey, objects near the detection threshold are preferentially scattered into the sample if they are intrinsically brighter (or scattered brighter) than average.

In the DESI TF survey, the initial parent sample is defined by the SGA catalog. We assume that SGA inclusion is complete above its angular-size cutoff and that, conditional on the TF model, the SGA selection is independent of the TF observables. In particular, there are no low surface-brightness galaxies that escape inclusion in the SGA catalog conditional on the angular size selection. Concretely, letting $S_{\mathrm{SGA}} \in \{0, 1\}$ denote SGA inclusion, we assume

$$p(\hat{x}_i, \hat{y}_i \mid \theta, S_{\mathrm{SGA},i} = 1) = p(\hat{x}_i, \hat{y}_i \mid \theta). \tag{A24}$$

Under this conditional-independence assumption, the SGA cutoff contributes only an overall constant to the likelihood and therefore does not affect TF parameter estimation.

The observed TF sample is then sculpted by DESI fiber allocation and subsequent redshift-measurement success. For the moment, we assume these processes are independent of galaxy properties, so they enter only through an overall completeness factor and do not need to be modeled explicitly in the likelihood. This assumption should be tested, since fiber allocation and redshift success can depend on quantities such as angular size, apparent magnitude and surface brightness.

We apply additional quality cuts to mitigate catastrophic outliers and to restrict the sample to galaxies for which the TF model is expected to be valid (e.g. excluding dwarf galaxies whose dynamics are not well described by the normal-galaxy TF relation). These cuts are modeled as hard selection on the measured observables: $(\hat{x}_i, \hat{y}_i) \in \mathcal{S}$, where $\mathcal{S}$ is the selection region. This is the sample selection considered in this section.

Finally, we apply a minimum redshift $z > z_{\min}$ to diminish the effect of peculiar velocities.

#### *b. Expanded Model and Likelihood*

With sample selection, we need to specify the model for the abundance of the parent galaxy population as a function of redshift (e.g. constant comoving number density). The expected number of sample galaxies, $\bar{N}$, is then calculable from the model and depends on the TF parameters. The realized sample size could then inform the inference of the TF parameters.

For simplicity and tractability, we do not introduce a model for the parent population abundance and instead treat the expected number of observed galaxies $\bar{N}$ as a free parameter in the likelihood: $\bar{N}$ does not depend on the TF parameters. In this case, the TF parameters are constrained only by the observed distribution of $(\hat{x}, \hat{y})$.

Assuming a Poisson model for the observed sample size and conditional independence of galaxy measurements, the total likelihood is

$$\mathcal{L}(\theta, \bar{N}; \{\hat{x}_i\}, \{\hat{y}_i\}) = \frac{e^{-\bar{N}} \bar{N}^N}{N!} \prod_{i=1}^{N} p(\hat{x}_i, \hat{y}_i \mid \theta, S_i = 1), \tag{A25}$$

where $S_i = 1$ denotes that galaxy $i$ is included by the selection.

For our choice of model, $\bar{N}$ is a dummy parameter that we profile over, so the relevant contribution to the likelihood is

$$\mathcal{L}(\theta; \{\hat{x}_i\}, \{\hat{y}_i\}) \propto \prod_{i=1}^{N} p(\hat{x}_i, \hat{y}_i \mid \theta, S_i = 1). \tag{A26}$$

For hard cuts applied to the *measured* observables, the selected-sample (truncated) per-galaxy density is

$$p(\hat{x}_i, \hat{y}_i \mid \theta, S_i = 1) = \frac{p(\hat{x}, \hat{y} \mid \theta)\, \mathbf{1}\{(\hat{x}, \hat{y}) \in \mathcal{S}\}}{P(S = 1 \mid \theta)} \tag{A27}$$

$$P(S_i = 1 \mid \theta) = \iint_{(\hat{x}, \hat{y}) \in \mathcal{S}} p(\hat{x}, \hat{y} \mid \theta)\, d\hat{x}\, d\hat{y}, \tag{A28}$$

where $\mathcal{L}_i$ is the per-galaxy likelihood of the complete-sample model computed above. Note that the $p(\hat{x}, \hat{y} \mid \theta) = \mathcal{L}_i$, the per-galaxy likelihood in the complete sample case (Eq. A8).

#### *c. Selection domain: lower and upper limits in $\hat{y}$ and a two-sided parallel half-plane cut*

We construct the sample by imposing four cuts: lower and upper limits on the measured absolute magnitude $\hat{y}$, and two parallel (oblique) half-plane cuts in the measured $(\hat{x}, \hat{y})$ plane. The upper $\hat{y}$ cut removes dwarf galaxies that do not follow the TF relation, while the oblique cuts remove extreme outliers observed in the DESI data.

We also impose a lower cut on $\hat{y}$ to reduce the dependence of the fit on the assumed form of the latent $y_{\mathrm{TF}}$ distribution. The top-hat and normal models differ primarily in their tail behavior while making similar predictions in the central region. Consequently, galaxies with very extreme $\hat{y}$ values—whose inferred $y_{\mathrm{TF}}$ would lie in the tails—can receive substantially different likelihoods under different choices of $p(y_{\mathrm{TF}})$. The $\hat{y}$ lower cut is therefore intended to remove these tail-dominated objects so that the inferred TF parameters are driven mainly by the region where the likelihood is comparatively insensitive to the specific $y_{\mathrm{TF}}$ model.

The four cuts can be written as the region $\mathcal{S} \subset \mathbb{R}^2$:

$$(\hat{x}, \hat{y}) \in \mathcal{S} \quad \Longleftrightarrow \quad \max\bigl\{\hat{y}_{\min},\ \bar{s}\,\hat{x} + \bar{c}_1\bigr\} \ \le\ \hat{y} \ \le\ \min\bigl\{\hat{y}_{\max},\ \bar{s}\,\hat{x} + \bar{c}_2\bigr\}. \tag{A29}$$

The parameters are chosen heuristically (by eye); fit results should be insensitive to small perturbations of their values.

Define the sheared coordinate

$$\hat{u} \equiv \hat{y} - \bar{s}\,\hat{x} - \bar{c}_1, \qquad d \equiv \bar{c}_2 - \bar{c}_1. \tag{A30}$$

In the sheared coordinates, collect the measured quantities into

$$\hat{\boldsymbol{z}}_i \equiv \begin{bmatrix} \hat{u}_i \\ \hat{y}_i \end{bmatrix}, \qquad \hat{u}_i \equiv \hat{y}_i - \bar{s}\,\hat{x}_i - \bar{c}_1. \tag{A31}$$

For fixed $y_{\rm TF}$,

$$\hat{\boldsymbol{z}}_i \mid y_{\rm TF} \sim \mathcal{N}_2\big(\boldsymbol{\mu}_i(y_{\rm TF}),\, \boldsymbol{\Sigma}_i\big), \tag{A32}$$

with

$$\boldsymbol{\mu}_i(y_{\rm TF}) = \begin{bmatrix} y_{\rm TF} - \bar{s}\,\dfrac{y_{\rm TF}-\beta}{\alpha} - \bar{c}_1 \\ y_{\rm TF} \end{bmatrix}, \qquad \boldsymbol{\Sigma}_i = \begin{bmatrix} \sigma_{u,i}^2 & \sigma_{2,i}^2 \\ \sigma_{2,i}^2 & \sigma_{2,i}^2 \end{bmatrix}, \qquad \sigma_{u,i}^2 \equiv \sigma_{2,i}^2 + \bar{s}^2\sigma_{1,i}^2. \tag{A33}$$

Define (for later convenience) the components

$$\mu_{u,i}(y_{\rm TF}) \equiv y_{\rm TF} - \bar{s}\,\frac{y_{\rm TF}-\beta}{\alpha} - \bar{c}_1, \qquad \mu_{y,i}(y_{\rm TF}) \equiv y_{\rm TF}. \tag{A34}$$

Define the (conditional) correlation

$$\rho_i \equiv \mathrm{Corr}(\hat{u}_i, \hat{y}_i \mid y_{\rm TF}) = \frac{\sigma_{2,i}^2}{\sigma_{u,i}\sigma_{2,i}} = \frac{\sigma_{2,i}}{\sigma_{u,i}}. \tag{A35}$$

With these definitions, the product of the 1D Gaussians appearing in the integrand of Eq. A14 can be written compactly as

$$\mathcal{N}\left(\hat{x}_i; \frac{y_{\rm TF}-\beta}{\alpha}, \sigma_{1,i}^2\right) \mathcal{N}\big(\hat{y}_i; y_{\rm TF}, \sigma_{2,i}^2\big) = |\bar{s}|\,\mathcal{N}_2(\hat{\boldsymbol{z}}_i; \boldsymbol{\mu}_i(y_{\rm TF}), \boldsymbol{\Sigma}_i)\,, \tag{A36}$$

where the factor $|\bar{s}|$ is the absolute Jacobian determinant of the linear change of variables $(\hat{x}, \hat{y}) \mapsto (\hat{u}, \hat{y})$. Therefore the per-galaxy marginal likelihood becomes

$$\mathcal{L}_i = |\bar{s}| \int dy_{\rm TF}\ \mathcal{N}_2(\hat{\boldsymbol{z}}_i; \boldsymbol{\mu}_i(y_{\rm TF}), \boldsymbol{\Sigma}_i)\ p(y_{\rm TF}). \tag{A37}$$

A feature of the sheared coordinates is that the lower oblique cut becomes $\hat{u} \geq 0$ and the upper oblique cut becomes $\hat{u} \leq d$. The $\hat{y}$ cuts remain axis-aligned, so in $(\hat{u}, \hat{y})$ coordinates the selection region is the rectangle

$$(\hat{u}, \hat{y}) \in \mathcal{S} \quad\Longleftrightarrow\quad 0 \leq \hat{u} \leq d, \quad \hat{y}_{\min} \leq \hat{y} \leq \hat{y}_{\max}. \tag{A38}$$

Because both $\hat{u}$ and $\hat{y}$ are constrained to finite intervals, the selection probability can be expressed via inclusion–exclusion as a difference of four quadrant integrals:

$$\begin{aligned} P(S_i = 1 \mid \theta) = {} & P(\hat{u}_i \geq 0,\ \hat{y}_i \leq \hat{y}_{\max} \mid \theta) - P(\hat{u}_i \geq d,\ \hat{y}_i \leq \hat{y}_{\max} \mid \theta) \\ & - P(\hat{u}_i \geq 0,\ \hat{y}_i \leq \hat{y}_{\min} \mid \theta) + P(\hat{u}_i \geq d,\ \hat{y}_i \leq \hat{y}_{\min} \mid \theta). \end{aligned} \tag{A39}$$

Writing $\Phi_2(h, k; \varrho)$ for the standard bivariate normal CDF with correlation $\varrho$, i.e. $\Phi_2(h, k; \varrho) = P(Z_1 \leq h, Z_2 \leq k)$, define the standardized bounds

$$h_i(t; y_{\rm TF}) \equiv \frac{\mu_{u,i}(y_{\rm TF}) - t}{\sigma_{u,i}}, \qquad k_i(y_\star; y_{\rm TF}) \equiv \frac{y_\star - \mu_{y,i}(y_{\rm TF})}{\sigma_{2,i}} = \frac{y_\star - y_{\rm TF}}{\sigma_{2,i}}. \tag{A40}$$

Then, for any $t \in \{0, d\}$ and $y_\star \in \{\hat{y}_{\min}, \hat{y}_{\max}\}$,

$$P(\hat{u}_i \geq t,\ \hat{y}_i \leq y_\star \mid \theta) = \int dy_{\rm TF}\ p(y_{\rm TF})\ \Phi_2\Big(h_i(t; y_{\rm TF}),\ k_i(y_\star; y_{\rm TF});\ -\rho_i\Big), \tag{A41}$$

where the sign flip in the correlation arises from rewriting $\hat{u}_i \geq t$ as $-\hat{u}_i \leq -t$. Substituting these quadrant terms into the inclusion–exclusion identity gives

$$\begin{aligned} P(S_i = 1 \mid \theta) = \int dy_{\rm TF}\ p(y_{\rm TF})\ \Big[ & \Phi_2\Big(h_i(0; y_{\rm TF}),\ k_i(\hat{y}_{\max}; y_{\rm TF});\ -\rho_i\Big) - \Phi_2\Big(h_i(d; y_{\rm TF}),\ k_i(\hat{y}_{\max}; y_{\rm TF});\ -\rho_i\Big) \\ & - \Phi_2\Big(h_i(0; y_{\rm TF}),\ k_i(\hat{y}_{\min}; y_{\rm TF});\ -\rho_i\Big) + \Phi_2\Big(h_i(d; y_{\rm TF}),\ k_i(\hat{y}_{\min}; y_{\rm TF});\ -\rho_i\Big)\Big]. \end{aligned} \tag{A42}$$

*d. Top-hat model for $y_{\mathrm{TF},i}$*

For the uniform prior

$$y_{\mathrm{TF}} \sim \mathrm{Uniform}(y_{\min}, y_{\max}), \qquad p(y_{\mathrm{TF}}) = \frac{1}{\Delta y_{\mathrm{TF}}}\,\mathbf{1}\{y_{\min} \le y_{\mathrm{TF}} \le y_{\max}\}\,, \qquad \Delta y_{\mathrm{TF}} \equiv y_{\max} - y_{\min},$$

the selection probability can be written as a 1D integral over the latent $y_{\mathrm{TF}}$:

$$P(S_i = 1 \mid \theta) = \frac{1}{\Delta y_{\mathrm{TF}}} \int_{y_{\min}}^{y_{\max}} I_i(y_{\mathrm{TF}})\, dy_{\mathrm{TF}}, \tag{A43}$$

where

$$\begin{aligned} I_i(y_{\mathrm{TF}}) \equiv\ & \Phi_2\Big(h_i(0; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\max}; y_{\mathrm{TF}});\ -\rho_i\Big) - \Phi_2\Big(h_i(d; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\max}; y_{\mathrm{TF}});\ -\rho_i\Big) \\ & - \Phi_2\Big(h_i(0; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\min}; y_{\mathrm{TF}});\ -\rho_i\Big) + \Phi_2\Big(h_i(d; y_{\mathrm{TF}}),\ k_i(\hat{y}_{\min}; y_{\mathrm{TF}});\ -\rho_i\Big). \end{aligned} \tag{A44}$$

As a function of $y_{\mathrm{TF}}$, the integrand $I_i(y_{\mathrm{TF}})$ is typically close to a top-hat: it transitions from $\approx 0$ to $\approx 1$ near $y_{\mathrm{TF}} \approx \hat{y}_{\min}$ and from $\approx 1$ to $\approx 0$ near $y_{\mathrm{TF}} \approx \hat{y}_{\max}$. For numerical integration it is therefore useful to concentrate quadrature points near both edges.

A convenient approach is to split the integral at a point $y_\star$ between the two transitions, e.g.

$$y_\star \equiv \min\Big\{y_{\max},\ \max\Big\{y_{\min},\ \frac{\hat{y}_{\min} + \hat{y}_{\max}}{2}\Big\}\Big\}, \qquad \int_{y_{\min}}^{y_{\max}} = \int_{y_{\min}}^{y_\star} + \int_{y_\star}^{y_{\max}}.$$

*a. Lower-edge reparameterization (near $\hat{y}_{\min}$).* Set

$$y_{\mathrm{TF}} = \hat{y}_{\min} + \sigma_{2,i} \sinh u_-, \qquad dy_{\mathrm{TF}} = \sigma_{2,i} \cosh u_-\, du_-,$$

with bounds

$$u_-^{\min} = \operatorname{asinh}\Big(\frac{y_{\min} - \hat{y}_{\min}}{\sigma_{2,i}}\Big), \qquad u_-^\star = \operatorname{asinh}\Big(\frac{y_\star - \hat{y}_{\min}}{\sigma_{2,i}}\Big).$$

Then

$$\int_{y_{\min}}^{y_\star} I_i(y_{\mathrm{TF}})\, dy_{\mathrm{TF}} = \sigma_{2,i} \int_{u_-^{\min}}^{u_-^\star} I_i\Big(\hat{y}_{\min} + \sigma_{2,i}\sinh u_-\Big) \cosh u_-\, du_-.$$

*b. Upper-edge reparameterization (near $\hat{y}_{\max}$).* Set

$$y_{\mathrm{TF}} = \hat{y}_{\max} - \sigma_{2,i} \sinh u_+, \qquad dy_{\mathrm{TF}} = -\sigma_{2,i} \cosh u_+\, du_+,$$

and flip the limits to keep increasing order. The corresponding bounds are

$$u_+^\star = \operatorname{asinh}\Big(\frac{\hat{y}_{\max} - y_\star}{\sigma_{2,i}}\Big), \qquad u_+^{\max} = \operatorname{asinh}\Big(\frac{\hat{y}_{\max} - y_{\max}}{\sigma_{2,i}}\Big),$$

so that

$$\int_{y_\star}^{y_{\max}} I_i(y_{\mathrm{TF}})\, dy_{\mathrm{TF}} = \sigma_{2,i} \int_{u_+^{\max}}^{u_+^\star} I_i\Big(\hat{y}_{\max} - \sigma_{2,i}\sinh u_+\Big) \cosh u_+\, du_+.$$

*c. Quadrature-friendly form.* Combining the two pieces,

$$P(S_i = 1 \mid \theta) = \frac{\sigma_{2,i}}{\Delta y_{\mathrm{TF}}} \left[\int_{u_-^{\min}}^{u_-^\star} I_i\Big(\hat{y}_{\min} + \sigma_{2,i}\sinh u_-\Big) \cosh u_-\, du_- + \int_{u_+^{\max}}^{u_+^\star} I_i\Big(\hat{y}_{\max} - \sigma_{2,i}\sinh u_+\Big) \cosh u_+\, du_+\right]. \tag{A45}$$

*e. Normal model for $y_{\mathrm{TF},i}$*

Assume

$$y_{\mathrm{TF}} \sim \mathcal{N}(\mu_{y_{\mathrm{TF}}}, \tau^2).$$

In this case, after marginalizing over $y_{\mathrm{TF}}$ the measured pair $(\hat{x}_i, \hat{y}_i)$ is jointly Gaussian (cf. the complete-sample result),

$$\begin{bmatrix}\hat{x}_i\\ \hat{y}_i\end{bmatrix} \sim \mathcal{N}_2\left(\begin{bmatrix}\mu_{x,i}\\ \mu_{y,i}\end{bmatrix}, \boldsymbol{\Sigma}_i^{(xy)}\right), \qquad \mu_{x,i} = \frac{\mu_{y_{\mathrm{TF}}} - \beta}{\alpha}, \quad \mu_{y,i} = \mu_{y_{\mathrm{TF}}}, \tag{A46}$$

with

$$\boldsymbol{\Sigma}_i^{(xy)} = \begin{bmatrix} \dfrac{\tau^2}{\alpha^2} + \sigma_{1,i}^2 & \dfrac{\tau^2}{\alpha} \\ \dfrac{\tau^2}{\alpha} & \tau^2 + \sigma_{2,i}^2 \end{bmatrix}, \qquad \alpha \neq 0. \tag{A47}$$

Introduce the sheared coordinate (as above)

$$\hat{u}_i \equiv \hat{y}_i - \bar{s}\,\hat{x}_i - \bar{c}_1, \qquad d \equiv \bar{c}_2 - \bar{c}_1, \qquad \hat{\boldsymbol{z}}_i \equiv \begin{bmatrix}\hat{u}_i\\ \hat{y}_i\end{bmatrix}.$$

Since $\hat{\boldsymbol{z}}_i$ is an affine transformation of a bivariate normal vector, it is also bivariate normal:

$$\hat{\boldsymbol{z}}_i \sim \mathcal{N}_2\left(\begin{bmatrix}\mu_{u,i}\\ \mu_{y,i}\end{bmatrix}, \boldsymbol{\Sigma}_i^{(uy)}\right), \tag{A48}$$

where

$$\mu_{u,i} = \mu_{y,i} - \bar{s}\,\mu_{x,i} - \bar{c}_1 = \mu_{y_{\mathrm{TF}}} - \bar{s}\,\frac{\mu_{y_{\mathrm{TF}}} - \beta}{\alpha} - \bar{c}_1, \tag{A49}$$

$$\sigma_{y,\mathrm{tot},i}^2 \equiv \Sigma_{yy,i}^{(uy)} = \tau^2 + \sigma_{2,i}^2, \tag{A50}$$

$$\Sigma_{uy,i}^{(uy)} = \sigma_{y,\mathrm{tot},i}^2 - \bar{s}\,\frac{\tau^2}{\alpha}, \tag{A51}$$

$$\sigma_{u,i}^2 \equiv \Sigma_{uu,i}^{(uy)} = \sigma_{y,\mathrm{tot},i}^2 + \bar{s}^2\left(\frac{\tau^2}{\alpha^2} + \sigma_{1,i}^2\right) - 2\bar{s}\,\frac{\tau^2}{\alpha}. \tag{A52}$$

Let

$$\rho_i^{(uy)} \equiv \mathrm{Corr}(\hat{u}_i, \hat{y}_i) = \frac{\Sigma_{uy,i}^{(uy)}}{\sigma_{u,i}\,\sigma_{y,\mathrm{tot},i}}, \qquad A_i(t) \equiv \frac{t - \mu_{u,i}}{\sigma_{u,i}}, \qquad B_i(y_\star) \equiv \frac{y_\star - \mu_{y,i}}{\sigma_{y,\mathrm{tot},i}}.$$

*a. Selection probability.* In $(\hat{u}, \hat{y})$ coordinates the selection region is the rectangle

$$(\hat{u}, \hat{y}) \in \mathcal{S} \iff 0 \le \hat{u} \le d,\ \hat{y}_{\min} \le \hat{y} \le \hat{y}_{\max}.$$

Therefore the selection probability is a bivariate-normal rectangle probability, which can be written by inclusion–exclusion in terms of the standard bivariate normal CDF $\Phi_2$:

$$\begin{aligned} P(S_i = 1 \mid \theta) &= P(0 \le \hat{u}_i \le d,\ \hat{y}_{\min} \le \hat{y}_i \le \hat{y}_{\max} \mid \theta) \\ &= \Phi_2\Big(A_i(d),\, B_i(\hat{y}_{\max});\, \rho_i^{(uy)}\Big) - \Phi_2\Big(A_i(0),\, B_i(\hat{y}_{\max});\, \rho_i^{(uy)}\Big) \\ &\quad - \Phi_2\Big(A_i(d),\, B_i(\hat{y}_{\min});\, \rho_i^{(uy)}\Big) + \Phi_2\Big(A_i(0),\, B_i(\hat{y}_{\min});\, \rho_i^{(uy)}\Big). \end{aligned} \tag{A53}$$

*b. Selected-sample per-galaxy density.* For galaxies that pass the cuts, the truncated likelihood contribution is

$$p(\hat{x}_i, \hat{y}_i \mid \theta, S_i = 1) = \frac{\mathcal{N}_2\left(\begin{bmatrix}\hat{x}_i\\ \hat{y}_i\end{bmatrix}; \begin{bmatrix}\mu_{x,i}\\ \mu_{y,i}\end{bmatrix}, \boldsymbol{\Sigma}_i^{(xy)}\right)}{P(S_i = 1 \mid \theta)}, \qquad (\hat{x}_i, \hat{y}_i) \in \mathcal{S}, \tag{A54}$$

with $P(S_i = 1 \mid \theta)$ given by Eq. (A53). (Equivalently, one may evaluate the numerator in $(\hat{u}, \hat{y})$ coordinates using $\mathcal{N}_2(\hat{\boldsymbol{z}}_i;\, [\mu_{u,i}, \mu_{y,i}]^T, \boldsymbol{\Sigma}_i^{(uy)})$ and the corresponding Jacobian.)

### 6. Priors

Use flat priors for $(\alpha, \beta)$ and weakly informative priors for the intrinsic scatters:

$$\sigma_{\mathrm{int},x} \sim \text{Half-Cauchy}(0, 0.3)\,, \tag{A55}$$
$$\sigma_{\mathrm{int},y} \sim \text{Half-Cauchy}(0, 0.3)\,. \tag{A56}$$

## Appendix B: Inferring the Absolute Magnitude of an Individual Galaxy from its Rotation Velocity

Assume we have already fit the TF model to a sample and obtained the posterior

$$p(\theta \mid \{\hat{x}_i, \hat{y}_i\}_{i=1}^{N}), \qquad \theta \equiv (\alpha, \beta, \sigma_{\mathrm{int},x}, \sigma_{\mathrm{int},y}, \mu_{\mathrm{TF}}, \tau). \tag{B1}$$

We now consider a *new* galaxy, denoted by "$\star$", for which we measure a rotation-velocity proxy $\hat{x}_\star$ with uncertainty $\sigma_{x,\star}$, and we wish to infer its true absolute magnitude $y_\star$ (or predict what absolute magnitude we should expect given $\hat{x}_\star$ and the TF calibration).

### 1. Posterior predictive distribution

The desired quantity is the posterior predictive distribution for a new galaxy with measured velocity proxy $\hat{x}_\star$ and uncertainty $\sigma_{x,\star}$:

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \{\hat{x}_i, \hat{y}_i\}) = \int d\theta\; p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta)\; p(\theta \mid \{\hat{x}_i, \hat{y}_i\})\,. \tag{B2}$$

Given posterior draws $\{\theta^{(m)}\}_{m=1}^{M}$, approximate this integral by

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \cdots) \;\approx\; \frac{1}{M}\sum_{m=1}^{M} p\Big(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta^{(m)}\Big)\,. \tag{B3}$$

*a. Single-draw predictive $p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta)$.* Introduce the latent Tully–Fisher magnitude $y_{\mathrm{TF},\star}$. Then

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \int dy_{\mathrm{TF},\star}\; p(y_\star \mid y_{\mathrm{TF},\star}, \theta)\; p(y_{\mathrm{TF},\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta)\,. \tag{B4}$$

Assume intrinsic scatter in $y$ at fixed $y_{\mathrm{TF}}$:

$$p(y_\star \mid y_{\mathrm{TF},\star}, \theta) = \mathcal{N}\big(y_\star \mid y_{\mathrm{TF},\star},\, \sigma^2_{\mathrm{int},y}\big)\,. \tag{B5}$$

To obtain $p(y_{\mathrm{TF},\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta)$, marginalize the latent true velocity proxy $x_\star$:

$$x_\star \mid y_{\mathrm{TF},\star}, \theta \sim \mathcal{N}\left(\frac{y_{\mathrm{TF},\star} - \beta}{\alpha},\, \sigma^2_{\mathrm{int},x}\right), \qquad \hat{x}_\star \mid x_\star \sim \mathcal{N}\big(x_\star,\, \sigma^2_{x,\star}\big)\,, \tag{B6}$$

which implies

$$\hat{x}_\star \mid y_{\mathrm{TF},\star}, \theta \sim \mathcal{N}\left(\frac{y_{\mathrm{TF},\star} - \beta}{\alpha},\, \sigma^2_{x,\mathrm{tot}}\right), \qquad \sigma^2_{x,\mathrm{tot}} \equiv \sigma^2_{\mathrm{int},x} + \sigma^2_{x,\star}. \tag{B7}$$

*b. Transformed Normal (likelihood as a Gaussian in $y_{\mathrm{TF},\star}$).* Define

$$\mu_L \equiv \beta + \alpha\hat{x}_\star, \qquad \sigma_L^2 \equiv \alpha^2\sigma^2_{x,\mathrm{tot}}, \qquad \sigma_L = |\alpha|\,\sigma_{x,\mathrm{tot}}. \tag{B8}$$

Then the Normal likelihood in $\hat{x}_\star$ can be rewritten exactly as

$$\mathcal{N}\left(\hat{x}_\star \,\middle|\, \frac{y_{\mathrm{TF},\star} - \beta}{\alpha},\, \sigma^2_{x,\mathrm{tot}}\right) = |\alpha|\,\mathcal{N}\big(y_{\mathrm{TF},\star} \mid \mu_L, \sigma_L^2\big)\,. \tag{B9}$$

Bayes' rule gives

$$p(y_{\rm TF,\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \frac{p(\hat{x}_\star \mid y_{\rm TF,\star}, \sigma_{x,\star}, \theta)\ p(y_{\rm TF,\star} \mid \theta)}{p(\hat{x}_\star \mid \sigma_{x,\star}, \theta)}$$
$$= \frac{\big[\,|\alpha|\,\mathcal{N}(y_{\rm TF,\star} \mid \mu_L, \sigma_L^2)\big]\ p(y_{\rm TF,\star} \mid \theta)}{|\alpha| \int dy\, \mathcal{N}(y \mid \mu_L, \sigma_L^2)\, p(y \mid \theta)}$$
$$= \frac{\mathcal{N}(y_{\rm TF,\star} \mid \mu_L, \sigma_L^2)\, p(y_{\rm TF,\star} \mid \theta)}{\int dy\, \mathcal{N}(y \mid \mu_L, \sigma_L^2)\, p(y \mid \theta)}, \tag{B10}$$

where the factor $|\alpha|$ cancels and $p(y \mid \theta)$ should be read as the prior for $y_{\rm TF}$.

*c. Case A: Gaussian model for $y_{\rm TF}$.* Assume

$$y_{\rm TF} \mid \theta \sim \mathcal{N}(\mu_{\rm TF}, \tau^2). \tag{B11}$$

**Denominator.** Using Eq. (B10), the denominator is the Gaussian–Gaussian convolution:

$$\int dy\, \mathcal{N}(y \mid \mu_L, \sigma_L^2)\, \mathcal{N}(y \mid \mu_{\rm TF}, \tau^2) = \mathcal{N}\big(\mu_L \mid \mu_{\rm TF},\ \sigma_L^2 + \tau^2\big)\,. \tag{B12}$$

Equivalently, in $\hat{x}_\star$-space,

$$p(\hat{x}_\star \mid \sigma_{x,\star}, \theta) = \mathcal{N}\left(\hat{x}_\star \,\middle|\, \frac{\mu_{\rm TF} - \beta}{\alpha},\ \sigma_{x,\rm tot}^2 + \frac{\tau^2}{\alpha^2}\right) = |\alpha|\, \mathcal{N}\big(\mu_L \mid \mu_{\rm TF},\ \sigma_L^2 + \tau^2\big)\,. \tag{B13}$$

**Posterior for $y_{\rm TF,\star}$.** From Eq. (B10),

$$p(y_{\rm TF,\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta) \propto \mathcal{N}(y_{\rm TF,\star} \mid \mu_L, \sigma_L^2)\, \mathcal{N}(y_{\rm TF,\star} \mid \mu_{\rm TF}, \tau^2), \tag{B14}$$

so

$$p(y_{\rm TF,\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \mathcal{N}(y_{\rm TF,\star} \mid \mu_\star,\ V_\star)\,, \tag{B15}$$

with

$$V_\star = \left(\frac{1}{\tau^2} + \frac{1}{\sigma_L^2}\right)^{-1}, \qquad \mu_\star = V_\star \left(\frac{\mu_{\rm TF}}{\tau^2} + \frac{\mu_L}{\sigma_L^2}\right). \tag{B16}$$

**Predictive for $y_\star$.** Using Eq. (B5),

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \mathcal{N}\big(y_\star \mid \mu_\star,\ \sigma_{{\rm int},y}^2 + V_\star\big)\,. \tag{B17}$$

*d. Case B: top-hat (uniform) model for $y_{\rm TF}$.* Assume a uniform prior on an interval:

$$p(y_{\rm TF} \mid \theta) = \mathrm{Unif}(y_{\min}, y_{\max}) = \frac{1}{y_{\max} - y_{\min}}\ \mathbf{1}_{[y_{\min}, y_{\max}]}(y_{\rm TF}). \tag{B18}$$

Let $\Phi(\cdot)$ and $\varphi(\cdot)$ denote the standard normal CDF and PDF.

**Denominator.** Using the transformed likelihood in Eq. (B9),

$$p(\hat{x}_\star \mid \sigma_{x,\star}, \theta) = \int dy\, \mathcal{N}\left(\hat{x}_\star \,\middle|\, \frac{y - \beta}{\alpha},\ \sigma_{x,\rm tot}^2\right)\ \frac{1}{y_{\max} - y_{\min}} \mathbf{1}_{[y_{\min}, y_{\max}]}(y)$$
$$= \frac{|\alpha|}{y_{\max} - y_{\min}} \int_{y_{\min}}^{y_{\max}} \mathcal{N}\big(y \mid \mu_L, \sigma_L^2\big)\ dy$$
$$= \frac{|\alpha|}{y_{\max} - y_{\min}} \left[\Phi\left(\frac{y_{\max} - \mu_L}{\sigma_L}\right) - \Phi\left(\frac{y_{\min} - \mu_L}{\sigma_L}\right)\right]. \tag{B19}$$

**Posterior for $y_{\mathrm{TF},\star}$.** From Eq. (B10), the posterior is proportional to $\mathcal{N}(y_{\mathrm{TF},\star} \mid \mu_L, \sigma_L^2)$ restricted to $[y_{\min}, y_{\max}]$; equivalently it is a truncated normal:

$$p(y_{\mathrm{TF},\star} \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \frac{\frac{1}{\sigma_L}\,\varphi\!\left(\frac{y_{\mathrm{TF},\star}-\mu_L}{\sigma_L}\right)\,\mathbf{1}_{[y_{\min},y_{\max}]}(y_{\mathrm{TF},\star})}{Z_L}, \qquad Z_L \equiv \Phi\!\left(\frac{y_{\max}-\mu_L}{\sigma_L}\right) - \Phi\!\left(\frac{y_{\min}-\mu_L}{\sigma_L}\right), \tag{B20}$$

where $\mu_L = \beta + \alpha\hat{x}_\star$ and $\sigma_L = |\alpha|\,\sigma_{x,\mathrm{tot}}$.

**Predictive for $y_\star$ (closed form).** Let $\sigma_y^2 \equiv \sigma_{\mathrm{int},y}^2$ and define

$$\sigma_Z^2 \equiv \sigma_L^2 + \sigma_y^2, \qquad \tilde{\sigma}^2 \equiv \left(\frac{1}{\sigma_L^2} + \frac{1}{\sigma_y^2}\right)^{-1}, \qquad \tilde{\mu}(y_\star) \equiv \tilde{\sigma}^2\left(\frac{\mu_L}{\sigma_L^2} + \frac{y_\star}{\sigma_y^2}\right). \tag{B21}$$

Then Eq. (B4) evaluates to

$$p(y_\star \mid \hat{x}_\star, \sigma_{x,\star}, \theta) = \mathcal{N}\big(y_\star \mid \mu_L,\ \sigma_Z^2\big)\ \frac{\Phi\!\left(\frac{y_{\max}-\tilde{\mu}(y_\star)}{\tilde{\sigma}}\right) - \Phi\!\left(\frac{y_{\min}-\tilde{\mu}(y_\star)}{\tilde{\sigma}}\right)}{Z_L}, \tag{B22}$$

with $Z_L$ from Eq. (B20).

## Appendix C: Color-Correction Model

### 1. Motivation

Empirical diagnostics of the TFR fit reveal color-dependent residuals: galaxies with redder $r-z$ colors are systematically brighter than predicted by the baseline model at fixed rotation velocity. We extend the model by introducing a single global parameter $\gamma$, the slope of the luminosity–color relation at fixed rotation velocity, so that a galaxy with $r-z$ color $c_i$ redder than the population mean $\mu_c$ has $r$-band absolute magnitude shifted by $\gamma\eta_i$, where $\eta_i$ is the component of color that is independent of velocity.

### 2. Generative Model

*a. Latent variables.* Each galaxy $i$ has latent variables $y_{\mathrm{TF},i}$, $x_i$, and $c_i$ (intrinsic color). The generative model is

$$y_{\mathrm{TF},i} \sim \mathrm{Uniform}(y_{\min},\, y_{\max}), \tag{C1}$$

$$x_i \mid y_{\mathrm{TF},i} \sim \mathcal{N}\!\left(\frac{y_{\mathrm{TF},i}-\beta}{\alpha},\ \sigma_{\mathrm{int},x}^2\right), \tag{C2}$$

$$c_i \mid x_i \sim \mathcal{N}\big(\mu_c + \delta(x_i - \bar{x}),\ \tau_c^2\big), \tag{C3}$$

$$\eta_i \equiv c_i - \mu_c - \delta(x_i - \bar{x}), \tag{C4}$$

$$y_i \mid y_{\mathrm{TF},i},\, \eta_i \sim \mathcal{N}\big(y_{\mathrm{TF},i} + \gamma\eta_i + \Delta_{r,i},\ \sigma_{\mathrm{int},y}^2\big), \tag{C5}$$

$$z_i \mid y_{\mathrm{TF},i},\, \eta_i,\, x_i \sim \mathcal{N}\big(y_{\mathrm{TF},i} + (\gamma-1)\eta_i - \mu_c - \delta(x_i - \bar{x}) + \Delta_{z,i},\ \sigma_{\mathrm{int},z}^2\big), \tag{C6}$$

with measurement equations

$$\hat{x}_i \mid x_i \sim \mathcal{N}(x_i,\ \sigma_{x,i}^2), \tag{C7}$$

$$\hat{y}_i \mid y_i \sim \mathcal{N}(y_i,\ \sigma_{y,i}^2), \tag{C8}$$

$$\hat{z}_i \mid z_i \sim \mathcal{N}(z_i,\ \sigma_{z,i}^2). \tag{C9}$$

Here $\bar{x}$ is the sample mean of $\hat{x}$ and

$$\Delta_{r,i} \equiv \alpha_{k,r}\big[\ln(1+z_{\mathrm{obs},i}) - \overline{\ln(1+z)}\big], \tag{C10}$$

$$\Delta_{z,i} \equiv \alpha_{k,z}\big[\ln(1+z_{\mathrm{obs},i}) - \overline{\ln(1+z)}\big] \tag{C11}$$

are band-dependent residual $k$-corrections that absorb any redshift-dependent offsets not already removed by the upstream photometric corrections, where $\overline{\ln(1+z)} \equiv N^{-1}\sum_j \ln(1+z_{\mathrm{obs},j})$ is the sample mean. Centering the regressors removes degeneracies with the intercept and $\mu_c$; the mean $k$-corrections are absorbed by these parameters by construction. Separate slopes are needed because the $r$- and $z$-band $k$-corrections differ: the observed $r-z$ color reddens by $\sim 0.12$ mag from $z_{\mathrm{obs}} = 0.02$ to 0.1 in the training sample, a trend that a single shared $\Delta_i$ cannot capture. The variable $\eta_i \sim \mathcal{N}(0, \tau_c^2)$ is the residual color after removing the population-level color–velocity slope $\delta$; it is independent of $x_i$ by construction.

*b. Physical interpretation.* The parameter $\gamma$ is the slope of the luminosity–color relation at fixed rotation velocity: at fixed $x_i$, a galaxy one unit redder in residual color ($\eta_i$ increased by 1) is $\gamma$ magnitudes brighter ($y_i$ more negative by $|\gamma|$ if $\gamma < 0$). The parameter $\delta$ captures the population-level trend that faster rotators tend to be redder; it is not a distance-indicator correction but rather a description of the galaxy population.

*c. Identification.* The model is identified because $\gamma$ enters only through the residual color $\eta_i$, which is orthogonal to $x_i$ by construction (Eq. C4). Specifically:

1. The transformation $c_i \to c_i + e$, $\beta \to \beta - \gamma e$ changes the prior mean of $c_i$ from $\mu_c$ to $\mu_c + e$, which is penalized by the prior $c_i \mid x_i \sim \mathcal{N}(\mu_c + \delta(x_i - \bar{x}), \tau_c^2)$ unless $\mu_c$ is simultaneously shifted. The prior $\mu_c \sim \mathcal{N}(\bar{c}_{\mathrm{obs}}, 1)$ (below) breaks this near-degeneracy.

2. The transformation $c_i \to c_i + d\,x_i$, $\alpha \to \alpha - \gamma d$ would require the color–velocity slope to shift by $d$, but $\gamma$ acts on $\eta_i$ (residual color at fixed $x_i$), not on $c_i$ directly; the linear-in-$x$ component of color is absorbed by $\delta$, not by $\alpha$. Because $\eta_i$ is defined using the latent $x_i$ rather than $\hat{x}_i$, no attenuation bias arises.

*d. Priors.* The parameters $\alpha$, $\beta$, $\sigma_{\mathrm{int},x}$, $\sigma_{\mathrm{int},y}$ inherit the priors of the baseline model (Appendix A). Additional priors:

$$\gamma \sim \mathcal{N}(0,\,1), \tag{C12}$$
$$\delta \sim \mathcal{N}(0,\,1), \tag{C13}$$
$$\mu_c \sim \mathcal{N}(\bar{c}_{\mathrm{obs}},\,1), \tag{C14}$$
$$\tau_c \sim \text{Half-Cauchy}(0,\,0.3) \text{ truncated to } \tau_c > 0.014, \tag{C15}$$
$$\ln\sigma_{\mathrm{int},z} \sim \mathcal{N}(-3,\,2), \tag{C16}$$
$$\alpha_{k,r} \sim \mathcal{N}(0,\,5), \tag{C17}$$
$$\alpha_{k,z} \sim \mathcal{N}(0,\,5), \tag{C18}$$

where $\bar{c}_{\mathrm{obs}} \equiv N^{-1}\sum_i(\hat{y}_i - \hat{z}_i)$ is the sample mean observed color. The prior on $\gamma$ is not truncated in the implementation. Without a hard constraint, the likelihood admits a near-degenerate solution at $\gamma > 0$ in which the roles of $\sigma_{\mathrm{int},y}$ and $\sigma_{\mathrm{int},z}$ are exchanged; fits to the data show that the $\gamma < 0$ mode has substantially higher posterior probability, and in practice the sampler concentrates on $\gamma < 0$ without needing the bound enforced explicitly. The lower bound $\tau_c > 0.014$ excludes a second degenerate mode at $\tau_c \approx 0.006$–$0.013$ in which the color correction effectively shuts off ($\gamma\tau_c \to 0$); this mode occupies a clearly separated region of the posterior with substantially lower probability than the physical mode at $\tau_c \approx 0.03$.

*e. Sampling parameterization.* Hamiltonian Monte Carlo on the natural $(\gamma, \tau_c)$ coordinates is hampered by two coupled features of the posterior geometry. First, the latent contribution to the $r$-band magnitude, $\gamma\eta_i \sim \mathcal{N}(0, p^2)$ with $p \equiv \gamma\tau_c$, depends on $\gamma$ and $\tau_c$ only through their product. The $y$-data alone are therefore degenerate along contours of constant $p$; the $z$- and color-data resolve $\gamma$ and $\tau_c$ separately only weakly, leaving a banana-shaped posterior ridge. Second, the prior on $\tau_c$ places appreciable mass near zero, and the joint posterior contracts sharply as $\tau_c \to 0$: a single HMC step size cannot simultaneously resolve the narrow region near $\tau_c = 0$ and the broader region away from it.

We mitigate both pathologies by sampling

$$(p,\,\ell) \;\equiv\; (\gamma\tau_c,\; \log\tau_c) \tag{C19}$$

in place of $(\gamma, \tau_c)$, with $\gamma = p\,e^{-\ell}$ and $\tau_c = e^{\ell}$ recovered as derived quantities. The $p$-coordinate aligns one sampler axis with the banana ridge; the log scale on $\tau_c$ renders the geometry scale-invariant near $\tau_c = 0$, removing the contraction. The change of variables has unit Jacobian determinant,

$$\left|\det\frac{\partial(p,\ell)}{\partial(\gamma,\tau_c)}\right| = \left|\det\begin{pmatrix}\tau_c & \gamma\\ 0 & \tau_c^{-1}\end{pmatrix}\right| = 1, \tag{C20}$$

because the $\tau_c^{-1}$ Jacobian from $\gamma \to p$ at fixed $\tau_c$ is exactly cancelled by the $\tau_c$ Jacobian from $\tau_c \to \ell$. The priors of Eqs. (C12)–(C18) therefore apply on the derived $\gamma$ and $\tau_c$ with no additional volume correction.

### 3. Marginalization over Latent Variables

*a. Marginalizing $\eta_i$.* For fixed $(y_{\mathrm{TF},i}, x_i)$, the residual color $\eta_i \sim \mathcal{N}(0, \tau_c^2)$ is independent of $x_i$. Since $y_i = y_{\mathrm{TF},i} + \gamma\eta_i + \Delta_{r,i} + \epsilon_y$ (with $\epsilon_y \sim \mathcal{N}(0, \sigma^2_{\mathrm{int},y})$) and $z_i = y_{\mathrm{TF},i} + (\gamma-1)\eta_i - \mu_c - \delta(x_i - \bar{x}) + \Delta_{z,i} + \epsilon_z$ (with $\epsilon_z \sim \mathcal{N}(0, \sigma^2_{\mathrm{int},z})$ independent of $\epsilon_y$), we have

$$\tilde{y}_i \equiv y_i - y_{\mathrm{TF},i} - \Delta_{r,i} = \gamma\eta_i + \epsilon_y, \tag{C21}$$
$$\tilde{z}_i \equiv z_i - [y_{\mathrm{TF},i} + \Delta_{z,i} - \mu_c - \delta(x_i - \bar{x})] = (\gamma-1)\eta_i + \epsilon_z. \tag{C22}$$

Marginalizing $\eta_i \sim \mathcal{N}(0, \tau_c^2)$ and adding measurement noise gives the joint marginal of $(\hat{y}_i, \hat{z}_i)$ given $(y_{\mathrm{TF},i}, x_i)$:

$$\begin{pmatrix} \hat{y}_i \\ \hat{z}_i \end{pmatrix} \Bigg| \, y_{\mathrm{TF},i},\, x_i \;\sim\; \mathcal{N}\left( \begin{pmatrix} y_{\mathrm{TF},i} + \Delta_{r,i} \\ y_{\mathrm{TF},i} + \Delta_{z,i} - \mu_c - \delta(x_i - \bar{x}) \end{pmatrix},\, \boldsymbol{A}_i \right), \tag{C23}$$

where the covariance matrix is

$$\boldsymbol{A}_i = \begin{pmatrix} \gamma^2\tau_c^2 + \sigma^2_{\mathrm{int},y} + \sigma^2_{y,i} & \gamma(\gamma-1)\tau_c^2 \\ \gamma(\gamma-1)\tau_c^2 & (\gamma-1)^2\tau_c^2 + \sigma^2_{\mathrm{int},z} + \sigma^2_{z,i} \end{pmatrix}. \tag{C24}$$

The off-diagonal $\gamma(\gamma-1)\tau_c^2$ arises because $\hat{y}$ and $\hat{z}$ share the color residual $\eta_i$ (with coefficients $\gamma$ and $\gamma - 1$); the intrinsic scatters $\epsilon_y$ and $\epsilon_z$ are independent and do not contribute.

*b. Sanity check: $\gamma = 0$, $\alpha_{k,r} = \alpha_{k,z} = 0$.* Setting $\gamma = 0$ and $\alpha_{k,r} = \alpha_{k,z} = 0$: $\boldsymbol{A}_i = \mathrm{diag}(\sigma^2_{\mathrm{int},y} + \sigma^2_{y,i},\ \sigma^2_{\mathrm{int},z} + \sigma^2_{z,i})$ and the off-diagonal vanishes, i.e., $\hat{y}$ and $\hat{z}$ are independent when color has no effect on $r$-band luminosity. The marginal for $\hat{y}_i$ reduces to $\mathcal{N}(y_{\mathrm{TF},i}, \sigma^2_{\mathrm{int},y} + \sigma^2_{y,i})$, recovering the baseline model of Appendix A.

*c. Marginalizing $x_i$.* Adding the measurement equation $\hat{x}_i \mid x_i \sim \mathcal{N}(x_i, \sigma^2_{x,i})$ and the model $x_i \mid y_{\mathrm{TF},i} \sim \mathcal{N}((y_{\mathrm{TF},i} - \beta)/\alpha,\ \sigma^2_{\mathrm{int},x})$, the variable $x_i$ enters the mean of $\hat{z}_i$ linearly through the $\delta(x_i - \bar{x})$ term. Marginalizing $x_i$ produces a trivariate Gaussian for $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ given $y_{\mathrm{TF},i}$. Defining

$$\sigma^2_{1,i} \equiv \sigma^2_{x,i} + \sigma^2_{\mathrm{int},x}, \tag{C25}$$
$$\mu_{x,i}(y_{\mathrm{TF}}) \equiv \frac{y_{\mathrm{TF}} - \beta}{\alpha}, \tag{C26}$$

the trivariate marginal is

$$\begin{pmatrix} \hat{x}_i \\ \hat{y}_i \\ \hat{z}_i \end{pmatrix} \Bigg| \, y_{\mathrm{TF},i} \;\sim\; \mathcal{N}\left( \begin{pmatrix} \mu_{x,i}(y_{\mathrm{TF},i}) \\ y_{\mathrm{TF},i} + \Delta_{r,i} \\ y_{\mathrm{TF},i} + \Delta_{z,i} - \mu_c - \delta(\mu_{x,i}(y_{\mathrm{TF},i}) - \bar{x}) \end{pmatrix},\, \boldsymbol{B}_i \right), \tag{C27}$$

where the $3 \times 3$ covariance matrix is

$$\boldsymbol{B}_i = \begin{pmatrix} \sigma^2_{1,i} & 0 & -\delta\sigma^2_{\mathrm{int},x} \\ 0 & A_{11} & A_{12} \\ -\delta\sigma^2_{\mathrm{int},x} & A_{12} & A_{22} + \delta^2\sigma^2_{\mathrm{int},x} \end{pmatrix}, \tag{C28}$$

with $A_{11}$, $A_{12}$, $A_{22}$ the entries of $\boldsymbol{A}_i$ (Eq. C24). The $(\hat{x}, \hat{y})$ off-diagonal is zero because $x_i$ does not enter $y_i$ once $y_{\mathrm{TF},i}$ is fixed; the $(\hat{x}, \hat{z})$ off-diagonal is $-\delta\sigma^2_{\mathrm{int},x}$ because $\hat{z}_i$ depends on the latent $x_i$ (not $\hat{x}_i$), and only the intrinsic variance $\sigma^2_{\mathrm{int},x}$ is shared between $\hat{x}_i$ and $\hat{z}_i$.

The remaining 1D integral over $y_{\mathrm{TF},i}$ admits a closed form because the mean vector in Eq. (C27) is linear in $y_{\mathrm{TF},i}$. Write

$$\boldsymbol{\mu}(y_{\mathrm{TF}}) = \boldsymbol{a}_i + \boldsymbol{b}\, y_{\mathrm{TF}}, \quad \boldsymbol{a}_i = \begin{pmatrix} -\beta/\alpha \\ \Delta_{r,i} \\ \Delta_{z,i} - \mu_c + \delta\bar{x} + \delta\beta/\alpha \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 1/\alpha \\ 1 \\ 1 - \delta/\alpha \end{pmatrix}. \tag{C29}$$

Let $\hat{\boldsymbol{d}}_i \equiv (\hat{x}_i, \hat{y}_i, \hat{z}_i)^T - \boldsymbol{a}_i$ and define the scalars

$$\xi_i \equiv \boldsymbol{b}^T \boldsymbol{B}_i^{-1} \boldsymbol{b}, \quad \phi_i \equiv \boldsymbol{b}^T \boldsymbol{B}_i^{-1} \hat{\boldsymbol{d}}_i, \quad \chi_i \equiv \hat{\boldsymbol{d}}_i^T \boldsymbol{B}_i^{-1} \hat{\boldsymbol{d}}_i. \tag{C30}$$

Completing the square in $y_{\rm TF}$,

$$(\hat{\boldsymbol{d}}_i - \boldsymbol{b}\, y_{\rm TF})^T \boldsymbol{B}_i^{-1} (\hat{\boldsymbol{d}}_i - \boldsymbol{b}\, y_{\rm TF}) = \xi_i\,(y_{\rm TF} - \mu_i^*)^2 + Q_{0,i}, \tag{C31}$$

with $\mu_i^* \equiv \phi_i/\xi_i$ and $Q_{0,i} \equiv \chi_i - \phi_i^2/\xi_i$. The integrand becomes a 1D Gaussian in $y_{\rm TF}$ times a $y_{\rm TF}$-independent prefactor; integrating over $[y_{\min}, y_{\max}]$ gives

$$\int_{y_{\min}}^{y_{\max}} \mathcal{N}_3\big((\hat{x}_i, \hat{y}_i, \hat{z}_i)^T;\, \boldsymbol{\mu}(y_{\rm TF}),\, \boldsymbol{B}_i\big)\, dy_{\rm TF} = \frac{e^{-Q_{0,i}/2}}{2\pi\sqrt{|\boldsymbol{B}_i|\,\xi_i}}\,\big[\Phi(u_i^{\max}) - \Phi(u_i^{\min})\big], \tag{C32}$$

where $u_i^\bullet \equiv \sqrt{\xi_i}\,(y_\bullet - \mu_i^*)$ and $\Phi$ is the standard normal CDF. Equation (C32) supersedes the sinh-reparameterization quadrature of Appendix A for the trivariate likelihood numerator; the selection integral in Section C 4 still requires 1D quadrature, since the denominator includes an additional bivariate integration over the selection region.

## 4. Selection Probability

The selection function $S_i = 1$ depends on the observed $(\hat{x}_i, \hat{y}_i)$ only; the $z$-band observation $\hat{z}_i$ is not used in the selection cuts (Eq. 10 of the main text). The selection probability is therefore

$$P(S_i = 1 \mid y_{{\rm TF},i},\, \theta,\, \gamma) = \iint_{(\hat{x},\hat{y})\in\mathcal{S}} p(\hat{x}_i, \hat{y}_i \mid y_{{\rm TF},i},\, \theta,\, \gamma)\, d\hat{x}_i\, d\hat{y}_i, \tag{C33}$$

where $p(\hat{x}_i, \hat{y}_i \mid y_{{\rm TF},i}, \theta)$ is the $(\hat{x}, \hat{y})$ marginal of Eq. (C27), i.e. a bivariate Gaussian with mean $(\mu_{x,i}(y_{{\rm TF},i}),\ y_{{\rm TF},i} + \Delta_{r,i})^T$ and covariance $\left(\begin{smallmatrix}\sigma_{1,i}^2 & 0\\ 0 & A_{11}\end{smallmatrix}\right)$. The $\hat{z}$ component does not enter because it is independent of $\hat{x}$ and $\hat{y}$ given $y_{{\rm TF},i}$; the mean shift from $\delta$ is carried only by $\hat{z}$, not by $\hat{y}$. Since $\Delta_{r,i}$ shifts $\mathrm{E}[\hat{y}_i \mid y_{{\rm TF},i}]$ by a galaxy-dependent amount, it enters the selection probability: at fixed $y_{{\rm TF},i}$, a non-zero $\Delta_{r,i}$ moves the predicted $\hat{y}_i$ relative to the fixed boundaries $\hat{y}_{\min}$, $\hat{y}_{\max}$. Equivalently, one may evaluate the selection integral with $\Delta_{r,i} = 0$ but with shifted boundaries $\hat{y}_{\min} - \Delta_{r,i}$ and $\hat{y}_{\max} - \Delta_{r,i}$. Substituting into the sheared-coordinate bivariate CDF machinery of Appendix A (Eq. A44), with $\sigma_{2,i}^2$ replaced by $A_{11} = \gamma^2\tau_c^2 + \sigma_{{\rm int},y}^2 + \sigma_{y,i}^2$, gives the selection probability as a 1D integral over $y_{{\rm TF},i}$.

## 5. Posterior Predictive Distribution

For a new galaxy with measurements $(\hat{x}_\star, \hat{z}_\star)$ and uncertainties $(\sigma_{x,\star}, \sigma_{z,\star})$, the posterior predictive for $\hat{y}_\star$ at fixed $(\theta, \gamma)$ is obtained by marginalizing $y_{{\rm TF},\star}$, $x_\star$, and $\eta_\star$.

*a. Marginalizing $\eta_\star$ and $x_\star$.* Given $y_{{\rm TF},\star}$, the trivariate marginal Eq. (C27) gives the joint distribution of $(\hat{x}_\star, \hat{y}_\star, \hat{z}_\star)$. Conditioning on the observed $(\hat{x}_\star, \hat{z}_\star)$ via the Schur complement gives

$$\hat{y}_\star \mid \hat{x}_\star,\, \hat{z}_\star,\, y_{{\rm TF},\star}\ \sim\ \mathcal{N}\big(\mu_{y|\hat{x}\hat{z}}(y_{{\rm TF},\star}),\ \sigma_{y|\hat{x}\hat{z}}^2\big), \tag{C34}$$

where

$$\mu_{y|\hat{x}\hat{z}}(y_{{\rm TF},\star}) = y_{{\rm TF},\star} + \Delta_{r,\star} + \boldsymbol{b}_{\rm reg}^T \begin{pmatrix} \hat{x}_\star - \mu_{x,\star}(y_{{\rm TF},\star}) \\ \hat{z}_\star - [y_{{\rm TF},\star} + \Delta_{z,\star} - \mu_c - \delta(\mu_{x,\star}(y_{{\rm TF},\star}) - \bar{x})] \end{pmatrix}, \tag{C35}$$

$$\sigma_{y|\hat{x}\hat{z}}^2 = A_{11} - \boldsymbol{b}_{\rm reg}^T \boldsymbol{D}_\star \boldsymbol{b}_{\rm reg}, \tag{C36}$$

with the regression coefficient vector

$$\boldsymbol{b}_{\rm reg} = \boldsymbol{D}_\star^{-1} \begin{pmatrix} 0 \\ A_{12} \end{pmatrix}, \qquad \boldsymbol{D}_\star = \begin{pmatrix} \sigma_{1,\star}^2 & -\delta\sigma_{{\rm int},x}^2 \\ -\delta\sigma_{{\rm int},x}^2 & A_{22} + \delta^2\sigma_{{\rm int},x}^2 \end{pmatrix}. \tag{C37}$$

The variance $\sigma_{y|\hat{x}\hat{z}}^2$ is independent of $y_{{\rm TF},\star}$.

*b. Marginalizing $y_{\mathrm{TF},\star}$.* The remaining integral over $y_{\mathrm{TF},\star}$ uses the joint posterior $p(y_{\mathrm{TF},\star} \mid \hat{x}_\star, \hat{z}_\star, \theta)$, which exploits the information that $\hat{z}_\star$ carries about $y_{\mathrm{TF},\star}$ through the dependence of $\hat{z}_\star$ on $y_{\mathrm{TF},\star}$. Marginalizing the trivariate Eq. (C27) over $\hat{y}_\star$ leaves a bivariate Gaussian likelihood

$$\begin{pmatrix} \hat{x}_\star \\ \hat{z}_\star \end{pmatrix} \Bigg| \, y_{\mathrm{TF},\star}, \theta \;\sim\; \mathcal{N}_2\big(\boldsymbol{m}_\star(y_{\mathrm{TF},\star}),\, \boldsymbol{D}_\star\big), \tag{C38}$$

with covariance $\boldsymbol{D}_\star$ as in Eq. (C37) and mean

$$\boldsymbol{m}_\star(y_{\mathrm{TF},\star}) = \boldsymbol{m}_\star(0) + y_{\mathrm{TF},\star}\, \boldsymbol{b}_{xz}, \qquad \boldsymbol{b}_{xz} = \begin{pmatrix} 1/\alpha \\ 1 - \delta/\alpha \end{pmatrix}. \tag{C39}$$

Combined with the uniform prior on $y_{\mathrm{TF},\star}$, the posterior is a truncated normal on $[y_{\min}, y_{\max}]$ whose untruncated precision and mean are

$$\xi_{xz} = \boldsymbol{b}_{xz}^T \boldsymbol{D}_\star^{-1} \boldsymbol{b}_{xz}, \qquad \mu_{xz}^\dagger = \xi_{xz}^{-1}\, \boldsymbol{b}_{xz}^T \boldsymbol{D}_\star^{-1} \big(\boldsymbol{o}_\star - \boldsymbol{m}_\star(0)\big), \tag{C40}$$

with $\boldsymbol{o}_\star = (\hat{x}_\star, \hat{z}_\star)^T$. Let $\mu_{T|xz}(\theta)$ and $V_{T|xz}(\theta)$ denote the mean and variance of this truncated normal (cf. Appendix B).

Because $\mu_{y|\hat{x}\hat{z}}(y_{\mathrm{TF},\star})$ is linear in $y_{\mathrm{TF},\star}$ with slope

$$s = 1 - \boldsymbol{b}_{\mathrm{reg}}^T \boldsymbol{b}_{xz}, \tag{C41}$$

the law of total expectation and variance give the exact conditional moments

$$\mathrm{E}[\hat{y}_\star \mid \hat{x}_\star,\, \hat{z}_\star,\, \theta,\, \gamma] = \mu_{y|\hat{x}\hat{z}}\big(\mu_{T|xz}(\theta)\big), \tag{C42}$$

$$\mathrm{Var}[\hat{y}_\star \mid \hat{x}_\star,\, \hat{z}_\star,\, \theta,\, \gamma] = \sigma^2_{y|\hat{x}\hat{z}} + s^2\, V_{T|xz}(\theta). \tag{C43}$$

*c. Posterior predictive over MCMC draws.* The posterior predictive mean and variance over draws $\{(\theta^{(m)}, \gamma^{(m)})\}_{m=1}^M$ are

$$\overline{y}_\star = \frac{1}{M} \sum_{m=1}^{M} \mathrm{E}^{(m)}[\hat{y}_\star \mid \hat{x}_\star, \hat{z}_\star, \theta^{(m)}, \gamma^{(m)}], \tag{C44}$$

$$\mathrm{Var}_{pp}(\hat{y}_\star) = \frac{1}{M} \sum_{m=1}^{M} \mathrm{Var}^{(m)} + \frac{1}{M} \sum_{m=1}^{M} \Big(\mathrm{E}^{(m)} - \overline{y}_\star\Big)^2. \tag{C45}$$

*d. Prediction from rotation velocity alone.* A reduced prediction that ignores $\hat{z}_\star$ is obtained by marginalizing the trivariate Eq. (C27) over $\hat{z}_\star$ rather than conditioning on it. The upper-left $2 \times 2$ block of $\boldsymbol{B}_\star$ (Eq. (C28)) has $B_{12} = 0$, so $\hat{x}_\star$ and $\hat{y}_\star$ are independent at fixed $y_{\mathrm{TF},\star}$ and

$$\hat{y}_\star \mid y_{\mathrm{TF},\star} \;\sim\; \mathcal{N}\big(y_{\mathrm{TF},\star} + \Delta_{r,\star},\; A_{11}\big), \tag{C46}$$

with $A_{11} = \gamma^2 \tau_c^2 + \sigma^2_{\mathrm{int},y} + \sigma^2_{y,\star}$ as before. The posterior of $y_{\mathrm{TF},\star}$ given $\hat{x}_\star$ alone is the truncated normal of the baseline model (Eq. (B20)), with mean $\mu_\star(\theta)$ and variance $V_\star(\theta)$. By the law of total expectation and variance,

$$\mathrm{E}[\hat{y}_\star \mid \hat{x}_\star,\, \theta,\, \gamma] = \mu_\star(\theta) + \Delta_{r,\star}, \tag{C47}$$

$$\mathrm{Var}[\hat{y}_\star \mid \hat{x}_\star,\, \theta,\, \gamma] = A_{11} + V_\star(\theta). \tag{C48}$$

Posterior-predictive moments over MCMC draws follow Eqs. (C44)–(C45) with $\mathrm{E}^{(m)}$ and $\mathrm{Var}^{(m)}$ replaced by the above. This $x$-only prediction is structurally identical to the baseline model (Appendix B) with $\sigma^2_{2,\star}$ replaced by $A_{11}$. Relative to the full $(\hat{x}_\star, \hat{z}_\star)$ prediction, it discards the $y_{\mathrm{TF},\star}$ information carried by $\hat{z}_\star$ but avoids propagating $z$-band photometric errors through $\boldsymbol{b}_{\mathrm{reg}}$; which prediction is tighter depends on the balance between these two effects.

### 6. Reduction to Baseline

Setting $\gamma = 0$ and $\alpha_{k,r} = \alpha_{k,z} = 0$ gives $A_{11} = \sigma^2_{\mathrm{int},y} + \sigma^2_{y,i} \equiv \sigma^2_{2,i}$, $A_{12} = 0$, $\Delta_{r,i} = \Delta_{z,i} = 0$, and $\boldsymbol{b}_{\mathrm{reg}} = \boldsymbol{0}$. The $\hat{z}$ measurement drops out of the prediction entirely: $\sigma^2_{y|\hat{x}\hat{z}} = A_{11} = \sigma^2_{2,i}$, recovering the baseline model of Appendix A. The likelihood at each galaxy reduces to the product $\mathcal{N}(\hat{x}_i; (y_{\mathrm{TF},i} - \beta)/\alpha, \sigma^2_{1,i})\, \mathcal{N}(\hat{y}_i; y_{\mathrm{TF},i}, \sigma^2_{2,i})\, \mathcal{N}(\hat{z}_i; y_{\mathrm{TF},i} - \mu_c - \delta(\mu_{x,i} - \bar{x}), \sigma^2_{\mathrm{int},z} + \sigma^2_{z,i})$, where the $\hat{z}$ factor is independent of $\hat{y}$ and contributes only to the estimation of $(\mu_c, \delta, \sigma_{\mathrm{int},z})$, not to the $r$-band prediction.

### 7. Model Limitations

The population color distribution is modeled as a single Gaussian $\mathcal{N}(\mu_c + \delta(x_i - \bar{x}), \tau_c^2)$. Non-Gaussian tails from dust, AGN contamination, or galaxy-type mixing are not captured; a mixture or nonparametric prior on $c_i$ is a natural extension. The independence of $\eta_i$ from $x_i$ (Eq. C4) is a model assumption equivalent to Gaussianity of the color–velocity residuals; it should be verified empirically against the observed $(\hat{c}_i, \hat{x}_i)$ scatter before the model is applied. The hyperparameter $\tau_c$ controls how much the $z$-band observation informs the latent $y_i$; naive estimation of $\tau_c$ from the observed color variance overestimates the intrinsic scatter because $\hat{c}_i = \hat{y}_i - \hat{z}_i$ has measurement noise $\sigma_{y,i}^2 + \sigma_{z,i}^2$; the posterior on $\tau_c$ from the full hierarchical fit automatically deconvolves this.

## Appendix D: Two-Color Extension

The single-color model of §C uses the $r-z$ color (one latent factor $\eta_i$) to sharpen the absolute-magnitude prediction. A natural extension adds $g$-band photometry, modeling the joint chromatic scatter of the three absolute magnitudes directly. This appendix describes the *2color* model, which replaces the trivariate likelihood $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ with a quadrivariate likelihood $(\hat{x}_i, \hat{y}_i, \hat{z}_i, \hat{g}_i)$.

### 1. Generative Model

Each galaxy retains the latent TF variables $y_{\mathrm{TF},i}$ and $x_i$ from §C 2. Rather than modeling the $r-z$ and $g-r$ colors through two *independent* latent color factors, the 2color model represents the joint intrinsic scatter of the three absolute magnitudes directly. Define the mean absolute-magnitude relations for the $r$, $z$, and $g$ bands,

$$m_{y,i} = y_{\mathrm{TF},i} + \Delta_{r,i}, \tag{D1}$$

$$m_{z,i} = y_{\mathrm{TF},i} - \mu_c - \delta_c(x_i - \bar{x}) + \Delta_{z,i}, \tag{D2}$$

$$m_{g,i} = y_{\mathrm{TF},i} - \mu_g - \delta_g(x_i - \bar{x}) + \Delta_{g,i}, \tag{D3}$$

where $\mu_c, \mu_g$ are the mean $r-z$ and $g-r$ colors at $x = \bar{x}$, $\delta_c, \delta_g$ the color–velocity slopes, and $\Delta_{r,i}, \Delta_{z,i}, \Delta_{g,i}$ the band residual $k$-corrections $[\Delta_{b,i} \equiv \alpha_{k,b}(\ln(1+z_{\mathrm{obs},i}) - \overline{\ln(1+z)})]$. The latent absolute magnitudes $(y_i, z_i, g_i)$ scatter about these means with a *rank-2* $3 \times 3$ intrinsic covariance,

$$\begin{pmatrix} y_i \\ z_i \\ g_i \end{pmatrix} \Bigg| \; y_{\mathrm{TF},i}, x_i \;\sim\; \mathcal{N}\left( \begin{pmatrix} m_{y,i} \\ m_{z,i} \\ m_{g,i} \end{pmatrix}, \boldsymbol{S} \right), \qquad \boldsymbol{S} = \boldsymbol{V}\,\boldsymbol{\Sigma}_c\,\boldsymbol{V}^T, \tag{D4}$$

where $\boldsymbol{V}$ is a $3 \times 2$ matrix with orthonormal columns spanning the plane orthogonal to a *free* unit null direction $\hat{\boldsymbol{n}}$, and $\boldsymbol{\Sigma}_c$ is a free $2 \times 2$ symmetric positive-definite matrix. [There should be a mention that runs with a rank-3 covariance showed near degeneracy, which led to slow convergence. This is why we go with rank-2.] By construction $\boldsymbol{S}\,\hat{\boldsymbol{n}} = \boldsymbol{0}$: $\boldsymbol{S}$ is rank 2, with a null space along $\hat{\boldsymbol{n}}$, so exactly one direction in $(y, z, g)$ space carries no intrinsic scatter — but *which* direction is inferred from the data, not fixed in advance. The unit vector $\hat{\boldsymbol{n}}$ is assigned a uniform prior on the sphere, and $\boldsymbol{V}$ is constructed as an orthonormal basis of its orthogonal complement (via a Householder reflector, so the map is smooth in $\hat{\boldsymbol{n}}$). The special case $\hat{\boldsymbol{n}} = \boldsymbol{e} = (1,1,1)/\sqrt{3}$ recovers the *chromatic-only* model, in which the excluded direction is exactly the achromatic (peculiar-velocity/distance-modulus) mode $(1,1,1)$ and all achromatic variance is deferred to the downstream velocity-covariance model $C_{vv}$ (§D 6); by fitting $\hat{\boldsymbol{n}}$ we instead let the data determine the excluded direction and *test* whether it coincides with $\boldsymbol{e}$ rather than assuming it does. The $2 \times 2$ covariance is parameterized as $\boldsymbol{\Sigma}_c = \boldsymbol{D}_c\,\boldsymbol{R}_c\,\boldsymbol{D}_c$ with $\boldsymbol{D}_c = \mathrm{diag}(S_{c,1}, S_{c,2})$ (two scatter scales) and $\boldsymbol{R}_c$ a $2 \times 2$ correlation matrix (one free parameter). The intrinsic scatter therefore has 5 free parameters in total — two for $\hat{\boldsymbol{n}}$, two scales, and one correlation — which is exactly the dimension of a general rank-2 positive-semidefinite $3 \times 3$ matrix; it sits strictly between the fixed-null chromatic-only form (3 parameters) and an unconstrained full-rank covariance (6). Only the plane $\mathrm{span}(\boldsymbol{V})$, equivalently $\hat{\boldsymbol{n}}$ up to sign, is identified: the in-plane orientation of the columns of $\boldsymbol{V}$ is a gauge freedom absorbed by the full $2 \times 2$ $\boldsymbol{\Sigma}_c$. Writing $S_{yy}, S_{yz}, S_{yg}, S_{zz}, S_{zg}, S_{gg}$ for the entries of $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{\Sigma}_c\boldsymbol{V}^T$, the measurement equations are $\hat{y}_i \mid y_i \sim \mathcal{N}(y_i, \sigma_{y,i}^2)$ and likewise for $\hat{z}_i, \hat{g}_i$, and the intrinsic (measurement-free) covariance of $(y_i, z_i, g_i)$ enters the likelihood through the $\boldsymbol{B}_i$ matrix below.

### 2. Marginalization to Quadrivariate Likelihood

Marginalizing $x_i$ over its intrinsic scatter $\sigma_{\mathrm{int},x}$ gives a quadrivariate Gaussian for $(\hat{x}_i, \hat{y}_i, \hat{z}_i, \hat{g}_i) \mid y_{\mathrm{TF},i}$ with mean vector

$$\boldsymbol{\mu}_i(y_{\mathrm{TF},i}) = \begin{pmatrix} \mu_{x,i}(y_{\mathrm{TF},i}) \\ y_{\mathrm{TF},i} + \Delta_{r,i} \\ y_{\mathrm{TF},i} + \Delta_{z,i} - \mu_c - \delta_c(\mu_{x,i} - \bar{x}) \\ y_{\mathrm{TF},i} + \Delta_{g,i} - \mu_g - \delta_g(\mu_{x,i} - \bar{x}) \end{pmatrix} \tag{D5}$$

and $4 \times 4$ covariance matrix

$$\boldsymbol{B}_i = \begin{pmatrix} \sigma_{1,i}^2 & 0 & -\delta_c \sigma_{\mathrm{int},x}^2 & -\delta_g \sigma_{\mathrm{int},x}^2 \\ 0 & A_{11} & A_{12} & A_{14} \\ -\delta_c \sigma_{\mathrm{int},x}^2 & A_{12} & A_{22} + \delta_c^2 \sigma_{\mathrm{int},x}^2 & A_{zg} \\ -\delta_g \sigma_{\mathrm{int},x}^2 & A_{14} & A_{zg} & A_{44} + \delta_g^2 \sigma_{\mathrm{int},x}^2 \end{pmatrix}, \tag{D6}$$

where the entries are set directly by the free intrinsic covariance $\boldsymbol{S}$ (Eq. D4),

$$A_{11} = S_{yy} + \sigma_{y,i}^2, \tag{D7}$$

$$A_{12} = S_{yz}, \tag{D8}$$

$$A_{14} = S_{yg}, \tag{D9}$$

$$A_{22} = S_{zz} + \sigma_{z,i}^2, \tag{D10}$$

$$A_{44} = S_{gg} + \sigma_{g,i}^2, \tag{D11}$$

$$A_{zg} = S_{zg} + \delta_c \delta_g \sigma_{\mathrm{int},x}^2. \tag{D12}$$

The $(3,4)$ entry has two contributions. $S_{zg}$ is a *free* parameter: the intrinsic $\hat{z}$–$\hat{g}$ covariance is no longer forced to zero, so $\hat{z}$ and $\hat{g}$ are not required to be conditionally independent given $(y_{\mathrm{TF},i}, x_i)$. Fixing $S_{zg} = 0$ recovers the earlier two-independent-latent-color-factor model. The second term is *induced* by the marginalization rather than free: because $m_{z,i}$ and $m_{g,i}$ both depend on $x_i$ with slopes $-\delta_c$ and $-\delta_g$ (Eqs. D4 ff.), integrating $x_i$ over its intrinsic scatter correlates $\hat{z}$ and $\hat{g}$ by $(-\delta_c)(-\delta_g)\sigma_{\mathrm{int},x}^2$, exactly as the same marginalization produces the $-\delta_c\sigma_{\mathrm{int},x}^2$ entries in the first row and column and the $\delta_c^2\sigma_{\mathrm{int},x}^2$, $\delta_g^2\sigma_{\mathrm{int},x}^2$ terms on the diagonal. Retaining it is what makes $\boldsymbol{B}_i$ the marginal covariance of the generative model, and hence positive definite by construction: $\boldsymbol{S}$ is positive semi-definite (rank 2, null along $\hat{\boldsymbol{n}}$) and the strictly positive per-band measurement variances $(\sigma_{y,i}^2, \sigma_{z,i}^2, \sigma_{g,i}^2)$ fill the null direction, guaranteeing that $\boldsymbol{B}_i$ is strictly positive definite. Dropping the induced term admits parameter values for which $\boldsymbol{B}_i$ has a negative eigenvalue and the Cholesky factorization fails.

### 3. Closed-Form Likelihood

The integral over $y_{\mathrm{TF},i}$ proceeds exactly as in §C 3: the mean $\boldsymbol{\mu}_i$ is affine in $y_{\mathrm{TF},i}$ with direction vector

$$\boldsymbol{b} = \begin{pmatrix} 1/\alpha \\ 1 \\ 1 - \delta_c/\alpha \\ 1 - \delta_g/\alpha \end{pmatrix}, \tag{D13}$$

so the closed-form log-likelihood is

$$\ln \mathcal{L}_i = -\tfrac{3}{2}\ln(2\pi) - \sum_k \ln L_{kk} - \tfrac{1}{2}\ln \xi_i - \tfrac{1}{2}Q_{0,i} + \ln \Delta\Phi_i - \ln(y_{\max} - y_{\min}), \tag{D14}$$

where $L_{kk}$ are the diagonal elements of the Cholesky factor of $\boldsymbol{B}_i$, and

$$\xi_i = \boldsymbol{b}^T \boldsymbol{B}_i^{-1} \boldsymbol{b}, \tag{D15}$$

$$\mu_i^\dagger = \phi_i/\xi_i, \quad \phi_i = \boldsymbol{b}^T \boldsymbol{B}_i^{-1} \boldsymbol{d}_i, \tag{D16}$$

$$Q_{0,i} = \boldsymbol{d}_i^T \boldsymbol{B}_i^{-1} \boldsymbol{d}_i - \phi_i^2/\xi_i, \tag{D17}$$

$$\Delta\Phi_i = \Phi\big(\sqrt{\xi_i}(y_{\max} - \mu_i^\dagger)\big) - \Phi\big(\sqrt{\xi_i}(y_{\min} - \mu_i^\dagger)\big), \tag{D18}$$

with $\boldsymbol{d}_i = \mathrm{obs}_i - \boldsymbol{\mu}_i(0)$. The normalization uses $-\frac{3}{2}\ln(2\pi)$ (rather than $-\frac{1}{2}\ln(2\pi)$ for the trivariate model) because the quadrivariate Gaussian has been reduced to a 1D integral; i.e., $4-1=3$ Gaussian dimensions are integrated analytically.

### 4. Posterior Predictive Distribution

Prediction proceeds as in §C 5 but with a $3\times 3$ conditioning matrix $\boldsymbol{D}_\star$ formed from the $(\hat{x},\hat{z},\hat{g})$ sub-block of $\boldsymbol{B}_\star$:

$$\boldsymbol{D}_\star = \begin{pmatrix} \sigma^2_{1,\star} & -\delta_c\sigma^2_{\mathrm{int},x} & -\delta_g\sigma^2_{\mathrm{int},x} \\ -\delta_c\sigma^2_{\mathrm{int},x} & A_{22}+\delta_c^2\sigma^2_{\mathrm{int},x} & A_{zg} \\ -\delta_g\sigma^2_{\mathrm{int},x} & A_{zg} & A_{44}+\delta_g^2\sigma^2_{\mathrm{int},x} \end{pmatrix}, \tag{D19}$$

with the $(\hat{z},\hat{g})$ entry $A_{zg} = S_{zg} + \delta_c\delta_g\sigma^2_{\mathrm{int},x}$ inherited directly from $\boldsymbol{B}_\star$ (Eq. D12): both the free intrinsic $S_{zg}$ and the term induced by marginalizing the shared latent $x_\star$ are retained, since $\boldsymbol{D}_\star$ is a genuine sub-block of $\boldsymbol{B}_\star$. Dropping the induced $\delta_c\delta_g\sigma^2_{\mathrm{int},x}$ term would make $\boldsymbol{D}_\star$ inconsistent with $\boldsymbol{B}_\star$ and forfeit the positive-definiteness it inherits from the latter. The regression coefficient vector is $\boldsymbol{b}_{\mathrm{reg}} = \boldsymbol{D}_\star^{-1}(0, A_{12}, A_{14})^T$ and the conditional moments are

$$\mu_{y|\hat{x}\hat{z}\hat{g}}(y_{\mathrm{TF},\star}) = y_{\mathrm{TF},\star} + \Delta_{r,\star} + \boldsymbol{b}_{\mathrm{reg}}^T\boldsymbol{r}_\star(y_{\mathrm{TF},\star}), \tag{D20}$$

$$\sigma^2_{y|\hat{x}\hat{z}\hat{g}} = A_{11} - (0, A_{12}, A_{14})\,\boldsymbol{D}_\star^{-1}\,(0, A_{12}, A_{14})^T, \tag{D21}$$

where $\boldsymbol{r}_\star$ is the 3-vector of residuals $(\hat{x}_\star - \mu_{x,\star},\ \hat{z}_\star - m_{z,\star},\ \hat{g}_\star - m_{g,\star})$ evaluated at $y_{\mathrm{TF},\star}$.

The posterior on $y_{\mathrm{TF},\star}$ given $(\hat{x}_\star, \hat{z}_\star, \hat{g}_\star)$ is a truncated normal with precision

$$\xi_{xzg} = \boldsymbol{b}_{xzg}^T\boldsymbol{D}_\star^{-1}\boldsymbol{b}_{xzg}, \qquad \boldsymbol{b}_{xzg} = \begin{pmatrix} 1/\alpha \\ 1-\delta_c/\alpha \\ 1-\delta_g/\alpha \end{pmatrix}, \tag{D22}$$

and mean $\mu^\dagger_{xzg} = \xi_{xzg}^{-1}\,\boldsymbol{b}_{xzg}^T\boldsymbol{D}_\star^{-1}(\boldsymbol{o}_\star - \boldsymbol{m}_\star(0))$. The exact posterior-predictive moments follow Eqs. (C42)–(C45) with $\boldsymbol{D}_\star$, $\boldsymbol{b}_{\mathrm{reg}}$, $\xi_{xzg}$, and $\boldsymbol{b}_{xzg}$ replacing their $2\times 2$ counterparts.

### 5. Prediction from Rotation Velocity Alone

Marginalizing $\hat{z}_\star$ and $\hat{g}_\star$ out of Eq. (D6) gives $\hat{y}_\star \mid y_{\mathrm{TF},\star} \sim \mathcal{N}(y_{\mathrm{TF},\star} + \Delta_{r,\star}, A_{11})$ (identical to Eq. (C46)), with $A_{11} = S_{yy} + \sigma^2_{y,\star}$ (Eq. D7); only the $y$-band intrinsic variance $S_{yy}$ enters, since $\hat{z}_\star$ and $\hat{g}_\star$ are marginalized out. The prediction follows Eqs. (C47)–(C48) with this $A_{11}$.

### 6. New Parameters and Priors

Relative to the single-color model, the 2color model adds the $g$-band mean structure $(\delta_g, \mu_g, \alpha_{k,g})$ and replaces the $r{-}z/g{-}r$ product-form intrinsic-scatter parameters $(\gamma, \tau_c, \gamma_g, \tau_g, \sigma_{\mathrm{int},y}, \sigma_{\mathrm{int},z}, \sigma_{\mathrm{int},g})$ with the rank-2 intrinsic covariance $\boldsymbol{S}$ of Eq. (D4), parameterized by the free null direction $\hat{\boldsymbol{n}}$, the two scatter scales $S_{c,1}, S_{c,2}$, and the $2\times 2$ correlation matrix $\boldsymbol{R}_c$:

| Parameter | Meaning | Prior |
|---|---|---|
| $\hat{\boldsymbol{n}}$ | rank-2 null direction | uniform on $S^2$ |
| $S_{c,1}, S_{c,2}$ | scatter scales | HalfCauchy$(0,1)$ |
| $\boldsymbol{R}_c$ | $2\times 2$ correlation | LKJ(2) |
| $\delta_g$ | population $g{-}r$ color–velocity slope | $\mathcal{N}(0,1)$ |
| $\mu_g$ | mean $g{-}r$ color at $x=\bar{x}$ | $\mathcal{N}(\bar{c}_{g,\mathrm{obs}}, 1)$ |
| $\alpha_{k,g}$ | $g$-band $k$-correction slope | $\mathcal{N}(0,5)$ |

The null direction $\hat{\boldsymbol{n}}$ is a unit vector with a uniform prior on the sphere, and $\boldsymbol{R}_c$ is sampled through its Cholesky factor (a single free element) with an LKJ(2) prior. The total parameter count for $\boldsymbol{S}$ is 5 (two for $\hat{\boldsymbol{n}}$, two scales, one correlation), between the 3 of the fixed-null chromatic-only special case and the 6 of an unconstrained free

$3 \times 3$ parameterization. The rank-2 restriction eliminates a near-degenerate ridge in the posterior that arises because photometry alone cannot distinguish intrinsic achromatic scatter from peculiar-velocity-induced scatter: with a free $3 \times 3$ $\boldsymbol{S}$, the fitted matrix is dominated by a single near-achromatic mode ($\sim 99\%$ correlated, $\sim 0.4$ mag) that is degenerate with $C_{vv}$, degrading HMC efficiency (mean treedepth at maximum, ESS ~0) and causing Cholesky failures on $\boldsymbol{B}_i$ when the sampler explores the boundary of the degenerate ridge. Restricting $\boldsymbol{S}$ to rank 2 places the sampler on a well-defined lower-dimensional manifold and removes this pathology, *without* hard-coding which direction is excluded: the null direction is fit, so the achromatic axis $\boldsymbol{e}$ is no longer imposed but can be compared against the posterior of $\hat{\boldsymbol{n}}$.

*a. Achromatic variance and the downstream velocity model.* The three absolute magnitudes share a common distance modulus: any peculiar-velocity error shifts $(y_i, z_i, g_i)$ by an identical amount along the achromatic direction $\boldsymbol{e} = (1,1,1)/\sqrt{3}$. The rank-2 model excludes exactly one direction from the intrinsic scatter ($\boldsymbol{S}\,\hat{\boldsymbol{n}} = \boldsymbol{0}$; Eq. D4), that direction being inferred. If the fitted $\hat{\boldsymbol{n}}$ coincides with $\boldsymbol{e}$, the excluded direction is precisely the peculiar-velocity/distance-modulus mode: the calibration model then contains *no* peculiar-velocity scatter, all achromatic variance is owned by the downstream velocity-covariance model $C_{vv}$ (which adds a coherent-flow prior to the per-galaxy distance-modulus posteriors of the prediction step, §D 4), and the separation is clean — $\boldsymbol{S}$ in the chromatic plane $\perp \boldsymbol{e}$, $C_{vv}$ along $\boldsymbol{e}$, no double-counting. We report the posterior angle between $\hat{\boldsymbol{n}}$ and $\boldsymbol{e}$ as a diagnostic; a posterior concentrated near $0^\circ$ *measures* the alignment that the fixed-null chromatic-only model would otherwise assume.

This design targets a specific risk: achromatic intrinsic scatter is only weakly identifiable from photometry alone — it is degenerate with peculiar-velocity-induced scatter — so fitting a full-rank $\boldsymbol{S}$ lets an achromatic component absorb an unknown fraction of the PV signal and bias the downstream flow-field estimate. The chromatic structure (intrinsic color scatter, color–velocity slope residuals), by contrast, is fully identifiable from the multi-band photometry and is precisely what the $g$-band extension is designed to measure. Because the excluded direction is weakly constrained, the posterior on $\hat{\boldsymbol{n}}$ may be broad; that is itself the honest outcome, and a posterior-predictive check on the color residuals ($z{-}y$, $g{-}y$) validates the chromatic structure independently of it.

### 7. Relation to the Single-Color Model

The rank-2 model nests the fixed-null chromatic-only model as the special case $\hat{\boldsymbol{n}} = \boldsymbol{e} = (1,1,1)/\sqrt{3}$, but for a generic fitted null direction it does *not* reduce to the single-color model of §C 3. The single-color model carries independent per-band intrinsic variances $(\sigma^2_{\mathrm{int},y}, \sigma^2_{\mathrm{int},z})$ that include achromatic scatter, whereas $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{\Sigma}_c\boldsymbol{V}^T$ is rank 2 and satisfies $\boldsymbol{S}\,\hat{\boldsymbol{n}} = \boldsymbol{0}$; it therefore cannot reproduce a full-rank (in particular, a generic diagonal) intrinsic covariance — a rank-2 matrix cannot be full rank, whatever $\hat{\boldsymbol{n}}$. The rank-2 restriction is a deliberate modeling choice (§D 6), not a special-case limit of the single-color model.

Setting the second scale $S_{c,2} = 0$ (so $\boldsymbol{\Sigma}_c$ is rank 1) makes $\boldsymbol{S}$ rank 1, $\boldsymbol{S} = S^2_{c,1}\,\boldsymbol{v}\boldsymbol{v}^T$ for the surviving column $\boldsymbol{v}$ of $\boldsymbol{V}$ (a unit vector in the plane $\perp \hat{\boldsymbol{n}}$); additionally taking $\delta_g = 0$ and $\alpha_{k,g} = 0$ makes $\hat{g}$ uninformative. This is the analog of the single-color model with a single intrinsic-scatter degree of freedom, but it still carries no scatter along $\hat{\boldsymbol{n}}$ — the independent $\sigma^2_{\mathrm{int},y}$ and $\sigma^2_{\mathrm{int},z}$ diagonal terms of the single-color $\boldsymbol{B}_i$ have no counterpart here.

## Appendix E: Redshift-Dependent Rotational-Velocity Correction

### 1. Motivation

Rotational velocities are measured at a fiber placed at $r_{\mathrm{fib}} = f\,R_{26}$, where $R_{26}$ is the $r$-band semi-major axis of the $SB = 26\ \mathrm{mag\,arcsec^{-2}}$ isophote and $f$ is the fiber fraction; for the DR2 measurements $f = 0.4$, corresponding to the tabulated $V_{0.4R_{26}}$. Because cosmological surface-brightness dimming scales as $(1+z)^4$, an isophote defined at a fixed *observed* surface brightness corresponds to a redshift-dependent *physical* radius. We therefore want every velocity to refer to the same physical location, $f\,R_{26}$ evaluated at a fiducial redshift $z_{\mathrm{fid}} = 0.05$. Left uncorrected, the drift of $R_{26}$ with redshift moves the fiber to a different physical radius as a function of $z$, imprinting a monotonic-in-$z$ systematic on the log-distance ratios that biases the $f\sigma_8$ inference.

The correction has three steps: (i) the surface-brightness shift of the isophote relative to $z_{\mathrm{fid}}$; (ii) the conversion of that shift to a fractional radius shift through the galaxy light profile; and (iii) the conversion of the radius shift to a velocity correction through the local slope of the rotation curve.

### 2. Surface-brightness shift of the isophote

Relative to the fiducial redshift, the observed surface brightness of a fixed physical isophote is shifted by

$$\delta\mu(z) = 10\,\log_{10}\frac{1+z}{1+z_{\rm fid}} + \big[\,k_\mu(z) - k_\mu(z_{\rm fid})\,\big], \tag{E1}$$

where the leading coefficient $10 = 2.5 \times 4$ is cosmological dimming and $k_\mu$ is the surface-brightness $k$-correction. In this work the SGA photometry is $K$-corrected to $z_{\rm fid}$ upstream and no separable per-galaxy $k_\mu$ is available, so we adopt $k_\mu(z) - k_\mu(z_{\rm fid}) \approx 0$ and retain only the dimming term (§E 6).

### 3. From surface brightness to radius via the light profile

We convert $\delta\mu$ to a radial shift using the SGA-2020 curve-of-growth fit [9],

$$m(r) = m_{\rm tot} + m_0\,\ln\!\big[\,1 + \alpha_1 (r/r_0)^{-\alpha_2}\,\big], \tag{E2}$$

with the scale radius fixed at $r_0 = 10''$. The surface brightness is the derivative of the enclosed flux per unit area; for elliptical isophotes of fixed axis ratio $q$ the enclosed area is $A(r) = \pi q\, r^2$, so with $F(<r) = 10^{-0.4\,m(r)}$ the local surface brightness is

$$\mu(r) = m(r) + 2.5\,\log_{10} r - 2.5\,\log_{10}\!\big[-m'(r)\big] + \text{const}, \tag{E3}$$

where the per-galaxy constant (axis ratio and zeropoints) is independent of $r$. The corrected isophote radius $r' = (1+\delta)\,R_{26}$ is the solution of

$$\mu(r') - \mu(R_{26}) = \delta\mu, \tag{E4}$$

which we evaluate directly from Eqs. (E2)–(E3). Because the difference depends only on the ratio $u(r)/u(R_{26}) = (r/R_{26})^{-\alpha_2}$, with $u \equiv \alpha_1 (r/r_0)^{-\alpha_2}$, the scale radius $r_0$ and the normalization $\alpha_1$ cancel, and $u(R_{26})$ is fixed by the measured aperture magnitude. The fractional shift is $\delta \equiv \delta r/R_{26} = r'/R_{26} - 1$; linearizing Eq. (E4) gives the closed form

$$\delta \simeq \frac{\delta\mu}{R_{26}\,(d\mu/dr)\big|_{R_{26}}}. \tag{E5}$$

For $z > z_{\rm fid}$ we have $\delta\mu > 0$: the dimmed isophote is observed at smaller radius, so the correction moves the target radius *outward*, $\delta > 0$.

### 4. From radius to velocity

A fractional change $\delta$ in the sampling radius maps to a velocity change through the local logarithmic slope of the rotation curve [19],

$$V_{\rm corr} = V_{\rm meas}\left[\,1 + \delta\,\frac{d\ln V}{d\ln r}\,\right]. \tag{E6}$$

Adopting the Ravi et al. [10] rotation-curve parameterization $V(r) = V_{\rm max}\, r\,/\,(R_{\rm turn}^{\alpha_{\rm RC}} + r^{\alpha_{\rm RC}})^{1/\alpha_{\rm RC}}$,

$$\frac{d\ln V}{d\ln r} = 1 - \frac{r^{\alpha_{\rm RC}}}{R_{\rm turn}^{\alpha_{\rm RC}} + r^{\alpha_{\rm RC}}}, \tag{E7}$$

where we hold the shape parameter fixed at $\alpha_{\rm RC} = 2.5$ as a provisional choice, carried as a symbol so the value is easy to revise. At the fiber $r = f\,R_{26}$ the ratio $r/R_{\rm turn} = f/(R_{\rm turn}/R_{26})$ is independent of $R_{26}$, so Eq. (E7) reduces to a single population value set by $R_{\rm turn}/R_{26}$.

### 5. Linearized correction and typical magnitude

Combining Eqs. (E1), (E5), and (E6),

$$\frac{V_{\rm corr}}{V_{\rm meas}} - 1 \simeq \frac{d\ln V}{d\ln r}\,\frac{10\,\log_{10}\big[(1+z)/(1+z_{\rm fid})\big]}{R_{26}\,\big(d\mu/dr\big)\big|_{R_{26}}}. \tag{E8}$$

For the DR2 sample the SGA light profile gives a median inverse slope $\big(R_{26}\,d\mu/dr\big)^{-1} \approx 0.19\ {\rm mag}^{-1}$, so $\delta \approx 0.076$ at $z = 0.15$. With the provisional $R_{\rm turn}/R_{26} = 0.25$ (giving $d\ln V/d\ln r \approx 0.24$), the velocity correction is $\approx +1.8\%$ at $z = 0.15$ relative to $z_{\rm fid} = 0.05$, rising to $\approx +3\%$ near $z \approx 0.25$ and vanishing by construction at $z = z_{\rm fid}$. The correction is small per galaxy but systematic in redshift.

### 6. Caveats

Several approximations enter the correction. (i) The log-slope $d\ln V/d\ln r$ is a population average; the Ravi et al. [10] MaNGA-matched rotation-curve fits are not yet incorporated, so the ratio $R_{\rm turn}/R_{26}$ (and hence the amplitude) is provisional and the correction scales linearly with it. (ii) The shape parameter $\alpha_{\rm RC} = 2.5$ is provisional. (iii) The radial conversion depends on the assumed SGA curve-of-growth light profile. The SGA $R_{26}$ isophotes are defined at a fixed *observed*-frame surface brightness, so the dimming correction is warranted; were they instead rest-frame or dimming-corrected the correction would not apply. (iv) The surface-brightness $k$-correction term $k_\mu$ is neglected. Although each per-galaxy correction is at the percent level, its systematic dependence on redshift makes it relevant for $f\sigma_8$ science.

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